

Field Guide to the Canadian Forest Fire Behavior Prediction (FBP) System

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Field Guide to the Canadian Forest Fire Behavior Prediction (FBP) System

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Disclaimer and limitation of liability

While care has been taken to ensure that the charts and tables in this guide accurately reflect the Canadian Forest Fire Behavior Prediction System (1992) for the stated conditions, no assurance is given that the work is free from error or omission. The Canadian Forest Service and the authors disclaim all liability for the consequences of anything done, or omitted to be done, by a person relying on the information given. Use of this guide is at the reader's sole risk and discretion.

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Introduction

The Canadian Forest Fire Behavior Prediction (FBP) System is a systematic method for assessing wildland fire behavior potential. It is a series of mathematical equations relating fire characteristics to wind, fuel moisture, and topographic conditions for 16 benchmark fuel (vegetation) types. Complete and precise fire behavior predictions are best made using computer-based representations of the system. However, this guide was prepared to allow first approximations of FBP System outputs when computer-based applications are not available. It is intended mainly to be used as a field reference and should be used in place of the forest fire growth calculator (McAlpine 1986), which is based on the interim version of the FBP System (Alexander et al. 1984). Certain simplifications were made so that important fire behavior characteristics could be presented in tabular form, and these are noted in the appropriate section. The symbols used in the guide are listed in Appendix 1 and many of the terms are defined in Appendix 2.

A complete technical description of the FBP System is given in Info. Rep. ST-X-3 Development and structure of the Canadian Forest Fire Behavior Prediction System (Forestry Canada Fire Danger Group 1992). Guidance on its application is given in the Canadian Forest Fire Behavior Prediction System User Guide (Hirsch 1996). Fire suppression interpretations based on head fire intensity class are also available for several fuel types (Alexander and De Groot 1988; Alexander and Lanoville 1989; Alexander and Cole 1995).

Caution

Fire behavior predictions are intended to assist in decision-making, and are not a substitute for experience, sound judgement, or observation of actual fire behavior. This document is not intended to be a guide to safe working conditions; almost any fire can be hazardous in some circumstances. Fire behavior can change rapidly due to changes in fuel conditions, slope, and exposure to wind. Because no system can ever fully account for all of the variables that affect fire behavior, operational personnel must be aware of the limitations of the system and be able to recognize unique or unusual situations.

Assumptions

Users must be careful not to apply the system beyond its useful range. In the FBP System, predictions are limited to a fire spreading during one burning period from a single point or line, assuming that:

- fuels conditions are similar to one of the 16 benchmark fuel types.
- the fuel moisture codes used are representative of the site conditions.
- fuels are uniform and continuous, topography is simple and homogeneous, and wind is constant and uni-directional.
- the fire is wind or wind/slope driven, and spread is not affected by a convection column. Wind is represented by the 10 m open wind.
- the fire is unaffected by suppression activities (free burning).
- a fire starting from a point source will have an elliptical shape under the above conditions.
- the effect of firebrands on spread is accounted for.

Use of the Guide

The following steps form the procedure for use of the guide:

1. Select the most appropriate **fuel type** (Table 1). In choosing a fuel model, the physical properties of the fuel complex such as surface fuel conditions, stand density, and crown-base height should be considered in addition to the tree species composition. Examples of the fuel types are given in Appendix 3.
2. Determine the most representative **Fine Fuel Moisture Code (FFMC)**. The **hourly FFMC** should be used, if available. Alternatively, the **diurnal FFMC** (Tables A.1-A.2, Appendix 5) may be used. The **daily standard FFMC** may be used to represent conditions at 1600 LST. The FFMC for slash and open fuel types may be further adjusted for topographic effects on clear days during the months of March, April, August, September, or October between 1200-2000 LST (Table A.3, Appendix 5).
3. Estimate the slope and determine the **slope equivalent wind speed** for the selected fuel type (Table 2). If the slope is $\geq 50\%$ and FFMC is ≥ 95 , add 5 to the table value.
4. Determine the on-site wind speed (measured, estimated, or forecast). Appendix 4 (Beaufort Scale) may be used as a guide if instruments are not available. If the wind is blowing upslope, **add** the observed wind speed to the slope equivalent wind speed. If the wind is blowing downslope, **subtract** the slope equivalent wind speed from the the observed wind speed. The result is the **effective wind speed**. Cross-slope fire spread results if the wind is blowing across the slope. However the cross-slope spread rate and direction are determined through vector addition, which is beyond the scope of this guide.
5. Find the **Initial Spread Index (ISI)** from FFMC and the effective wind speed (Table 3). Also, note the back fire initial spread index (**BISI**) if greater than zero.
6. Determine the **Buildup Index (BUI)** for the prediction point from the closest representative fire weather station. Find the **equilibrium head fire rate of spread (ROS)** and the **back fire rate of spread (BROS)** for the selected fuel type (Tables 4.1-4.26) from the ISI/BISI and BUI. The **type of fire** is indicated in the tables as follows:

black figures	surface fire	< 10 % crown fraction burned
black figures with *	intermittent crown fire	10 - 89 % crown fraction burned
white figures	continuous crown fire	>90 % crown fraction burned

The line dividing the intermittent crown fire class is the 50% crown fraction burned (CFB) level. The background shading in the charts indicates the fire intensity class. BUI_0 (the BUI value that has no effect on ROS) is also indicated and can be used if the BUI is not known.

7. **Head fire spread distance** and **back fire spread distance** for various prediction intervals can be determined from the equilibrium ROS or BROS using Table 5.1 or 5.2. Distances are given for three spread functions:

Equilibrium ROS in all fuel types

Accelerating ROS in open fuel types and surface fires in closed fuel types

Accelerating ROS in crown fires in closed fuel types

Use the spread values based on the equilibrium ROS for fires spreading from an active perimeter. Use the spread values with an acceleration adjustment for fires spreading from a point source.

8. **Fire size** and **perimeter length** of a fire starting from a point source can be estimated for a given time period from the total spread distance and effective wind speed (Table 6.1 or 6.2). The **perimeter growth rate** can be determined from ROS and effective wind speed in Table 7.1 or 7.2. Values are given for both timber/slash and grass fuel types. Fire width can be estimated by dividing the total spread distance by the length/breadth (L/B) ratio (Table 6.1 or 6.2).

Accuracy of the Guide

This guide will give a good estimate of the Fire Intensity Class and Type of Fire in most cases. In most fuel types users should be able to estimate fire size characteristics within $\pm 20\%$ of computed values for $ROS > 3$.

Fire characteristics are less accurate for the 0-20 BUI class and for $ROS \leq 3$ in because of interpolation and rounding errors.

Procedures used in the guide

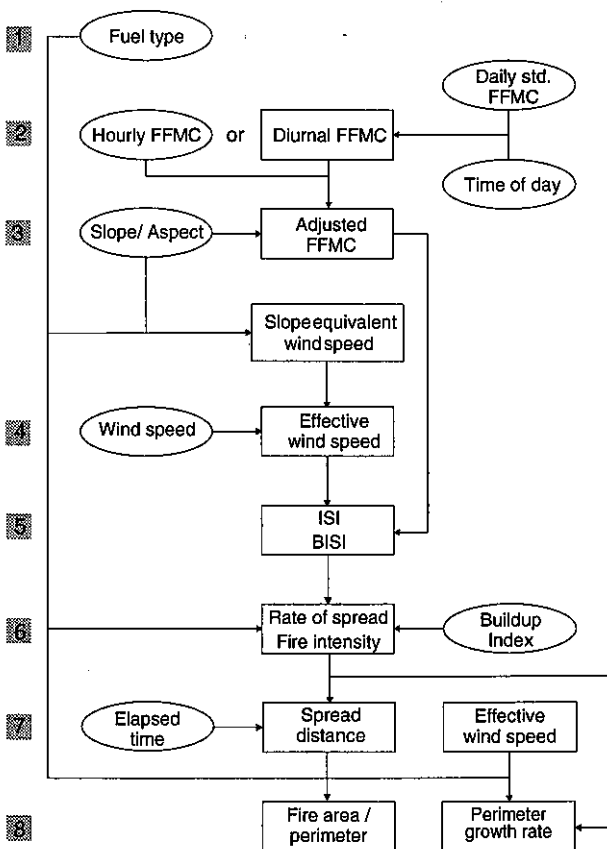


TABLE 1
FBP System
Fuel types

Group	Identifier	Description
Coniferous	C-1	Spruce-lichen woodland
	C-2	Boreal spruce
	C-3	Mature jack or lodgepole pine
	C-4	Immature jack or lodgepole pine
	C-5	Red and white pine
	C-6	Conifer plantation
	C-7	Ponderosa pine-Douglas-fir
Deciduous	D-1	Leafless aspen
Mixedwood*	M-1	Boreal mixedwood-leafless
	M-2	Boreal mixedwood-green
	M-3	Dead balsam fir mixedwood-leafless
	M-4	Dead balsam fir mixedwood-green
Slash	S-1	Jack or lodgepole pine slash
	S-2	White spruce-balsam slash
	S-3	Coastal cedar-hemlock-Douglas-fir slash
Open	O-1a	Matted grass
	O-1b	Standing grass

* M-1 and M-2 are transitional between C-2 and D-1

TABLE 2

Slope equivalent wind speed (km/h)

Fuel type identifier	Slope (%)					
	10	20	30	40	50	60
C-1	1	3	4	6	8	10
C-2	3	7	12	17	23	29
C-3	2	4	6	9	12	15
C-4	3	7	12	17	23	29
C-5	1	3	5	7	9	12
C-6	2	4	7	10	13	17
C-7	2	5	9	13	17	21
D-1	3	7	11	16	21	26
M-1	3	7	12	16	22	28
M-2	3	7	11	16	21	26
M-3 30PDF	3	7	11	16	21	26
60PDF	3	8	13	18	24	32
100PDF	4	9	14	21	29	41
M-4 30PDF	2	5	8	11	14	18
60PDF	2	6	9	13	18	23
100PDF	3	8	13	18	25	32
S-1	4	8	14	20	27	35
S-2	3	7	11	16	21	26
S-3	2	4	6	9	12	16
O-1a	3	8	13	19	25	32
O-1b	3	7	11	15	21	26

Find the fuel type identifier, then move horizontally to the most appropriate slope column to read the **slope equivalent wind speed**. If the FFMC ≥ 95 and slope $\geq 50\%$ add 5 to the table value.

Slope equivalent windspeed varies with FFMC. Those given above are for FFMC 90 and are accurate to ± 2 km/h of the true values for FFMC 80-96. The values for FFMC ≥ 95 and slope $\geq 50\%$ may be under estimated by ≥ 5 km/h.

M-1 and M-2 values are for 50:50 conifer / hardwood. M-3 and M-4 modifiers are percent dead fir (PDF).

TABLE 3

ISI/BISI Initial Spread Index

Back fire Initial Spread Index

Effective wind speed (km/h)

FFMC	0	5	10	15	20	25	30	35	40	45	50
77	1	1	1	2	2	3	4	5	7	8	9
78	1	1	1	0	0	0	0	0	0	0	0
79	1	1	2	2	3	4	5	6	8	9	10
80	1	1	1	1	0	0	0	0	0	0	0
81	1	2	2	3	3	4	6	7	10	11	13
82	1	2	2	3	4	5	6	8	11	13	14
83	2	2	3	3	4	6	7	9	12	15	16
84	2	2	3	4	5	6	8	11	14	17	18
85	2	3	3	4	6	7	10	12	16	19	21
86	2	3	4	5	7	9	11	14	18	22	24
87	3	4	5	6	8	10	13	16	21	25	28
88	3	4	5	7	9	11	15	19	24	29	32
89	4	5	6	8	10	13	17	22	28	33	37
90	4	6	7	9	12	15	19	25	32	39	43
91	5	6	8	11	14	17	22	29	37	45	50
92	6	7	9	12	16	20	26	33	43	51	57
93	7	8	11	14	18	23	30	38	49	59	66
94	8	10	12	16	21	27	34	44	57	68	76
95	9	11	14	18	24	31	39	51	65	78	87
96	10	13	16	21	27	35	45	58	74	89	99
97	11	15	19	24	31	40	51	66	85	102	114
98	13	17	21	28	36	46	59	76	97	117	130

Red figures are ISI

Black figures are BISI

TABLE 4.1
Equilibrium rate of spread (m/min)
and fire intensity class

C-1 Spruce-lichen woodland

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

BUI

ISI	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
-----	---------	----------	----------	----------	----------	-----------	------------	------------

1					1			
2								
3								
4								
5					2			
6								
7								
8	1	1	1	1	2	2	2	2
9	1	2*	2*	2*	3	2*	2*	2*
10	2*	3*	3*	3*	3*	3*	3*	3*
11	3*	4*	4*	4*	4	4*	5*	5*
12	4*	5*	5*	5*	6*	6*	6*	6*
13	5*	6*	7*	7*	7*	7*	7*	7*
14	6*	8*	8*	9*	9*	9*	9*	9*
15	7*	9*	10*	10*	5	11*	11*	11*
16	8*	11*	12	12	13	13	13	13
17	9*	13	14	14	15	15	15	15
18	11*	15	16	16	17	17	17	18
19	12	17	18	19	19	20	20	20
20	14	19	20	21	21	22	22	22
21-25	18	25	27	28	29	29	30	30
26-30	26	35	38	39	40	41	42	42
31-35	33	45	48	50	51	52	53	54
36-40	39	53	56	59	6	60	62	63
41-45	44	59	63	66	68	69	70	71
46-50	50	64	68	71	73	75	76	77
51-55	51	68	72	75	78	79	81	81
56-60	52	71	75	78	81	83	84	85
61-65	54	73	77	81	83	85	86	87
66-70	55	75	79	83	85	87	88	89

Constants: foliar moisture content= 97%; CBH = 2m; surface fuel consumption for FPMC 90

Type of fire: **surface** **intermittent crown*** **continuous crown** _ CFB 50% **BUI₀**

TABLE 4.2

Equilibrium rate of spread (m/min)
and fire intensity class**C-2 boreal spruce**

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

ISI	BUI							
	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
1	0.4*	0.4	0.4	0.4	2	0.6	0.6	0.6
2	0.5	0.5	1	1	3	1	2*	2*
3	0.5	2	2	2*	3*	3*	3*	3*
4	0.5	3	3*	4*	4*	4*	4*	5*
5	0.5	3*	4*	5*	5*	6*	6*	6*
6	0.5	4*	5*	6*	5	7*	7*	8*
7	0.5	5*	7*	8*	9*	9*	10*	10*
8	0.5	7*	8*	9*	10*	11	12	12
9	0.5	8*	9*	11*	12	13	14	14
10	3	9*	11*	12	14	15	16	16
11	4	10*	12	14	16	17	18	18
12	4	11*	14	16	17	19	20	20
13	4	12	15	17	19	21	22	22
14	5	13	16	19	21	23	24	25
15	5	15	18	21	6	23	25	26
16	6*	16	19	22	25	27	28	29
17	6*	17	21	24	27	29	30	31
18	6*	18	22	26	28	31	32	33
19	7*	19	23	27	30	33	34	35
20	7*	20	25	29	32	34	36	37
21-25	8*	24	29	34	37	40	42	43
26-30	10*	29	35	41	46	49	52	53
31-35	12*	34	41	48	53	57	60	62
36-40	14*	39	47	54	60	65	68	70
41-45	15	43	52	60	66	72	75	78
46-50	16	46	56	65	72	78	82	84
51-55	17	49	60	70	77	83	87	90
56-60	18	52	64	74	82	88	92	95
61-65	19	55	67	77	85	92	97	100
66-70	20	57	69	80	89	96	101	104

Constants: foliar moisture content = 97% ; crown base height = 3m

Type of fire: surface intermittent crown* continuous crown _ CFB 50% BUI₀

TABLE 4.3

Equilibrium rate of spread (m/min)
and fire intensity class**C-3 mature jack or lodgepole pine**

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

BUI

	0	21	31	41	61	81	121	161
ISI	20	30	40	60	80	120	160	200
1					1			
2								
3					2			
4							0.5	0.5
5					0.9	0.9	1	1
6	0.4	1	1	1	1	2	2	2
7	0.7	2	2	2	3	2	2	2
8	1	3	2	3	3	3	3	3*
9	1	3	3	4	4	4	4*	5*
10	2	4	4	5	5	6*	6*	6*
11	3	5	5	6	7*	7*	7*	7*
12	4	6	7	7	8*	8*	9*	9*
13	5	7	8	9	10*	10*	11*	11*
14	6	8	9	10*	11*	12*	12*	13*
15	7	9	11	12*	13*	14*	14	15
16	8	10	12	14*	15*	16	16	17
17	9	12	14	16*	17	18	19	19
18	10	13	15	17*	19	20	21	21
19	11	15	17*	19	21	22	23	24
20	12	16	19*	21	23	25	26	26
21-25	13	21	24*	27	30	31	33	34
26-30	14	28*	33	38	41	43	45	46
31-35	15	36*	42	47	51	55	57	58
36-40	16	43	50	56	61	65	68	69
41-45	17	49	57	64	70	74	77	79
46-50	23	54	63	71	77	82	86	88
51-55	25	58	69	77	84	89	93	95
56-60	27	62	73	82	89	95	99	101
61-65	28	65	77	86	94	100	104	106
66-70	29	68	80	90	97	103	108	110

Constants: foliar moisture content = 97% ; crown base height = 8m

Type of fire: **surface** **intermittent crown*** **continuous crown** _ CFB 50% **BUI₀**

TABLE 4.4

Equilibrium rate of spread (m/min)
and fire intensity class

C-4 immature jack or lodgepole pine

INTENSITY CLASS	
1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

ISI	BUI							
	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
1	1	1	1	1	1	0.6	0.6	0.6
2	1	1	1	1	2	2	2*	2*
3	1	2	2	3	3*	3*	3*	3*
4	2	3	3	4*	4*	4*	4*	5*
5	3	4	5	5*	6*	6*	6*	6*
6	4	5	6*	7*	7*	8*	8*	8*
7	5	7	8*	8*	9*	9*	10*	10*
8	6	8	9*	10*	11*	11*	11	12
9	7	9	11*	12*	12	13	13	14
10	8	11*	12*	13	14	15	15	16
11	9	12*	14*	15	16	17	17	18
12	10	14*	15*	17	18	19	19	20
13	11	15*	17	19	20	21	21	22
14	12	16*	19	20	22	23	23	24
15	13	18*	20	22	24	25	26	26
16	14	19*	22	24	25	27	28	28
17	15	21	23	26	27	29	30	30
18	16	22	25	27	29	31	32	32
19	17	23	27	29	31	33	34	34
20	18	25	28	31	33	34	36	36
21-25	19	29	33	36	38	40	41	42
26-30	20	35	40	44	47	49	50	51
31-35	21	41	46	51	54	57	59	60
36-40	24	46	52	57	61	64	66	67
41-45	27	51	58	63	67	71	73	74
46-50	29	55	62	69	73	76	79	80
51-55	31	59	67	73	78	82	84	86
56-60	32	62	70	77	82	86	89	90
61-65	34	65	74	81	86	90	93	95
66-70	35	67	76	84	89	94	97	98

Constants: foliar moisture content = 97%; crown base height = 4m

Type of fire: **surface** **intermittent crown*** **continuous crown** _ CFB 50% [BUI₀]

TABLE 4.5
Equilibrium rate of spread (m/min)
and fire intensity class

C-5 red and white pine

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

	BUI							
	0	21	31	41	61	81	121	161
ISI	20	30	40	60	80	120	160	200
1					1			
2								
3								
4								
5				2				
6								0.5
7						0.7	0.8	0.8
8					1	1	1	1
9		1	1		1	2	2	2
10		2	2	2	2	2	2	2
11	1	2	2	3	3	3	3	3
12	1	3	3	3	3	3	3	4
13	2	3	3	4	4	4	4	4
14	2	4	4	4	5	5	5	5
15	3	4	5	5	6	6	6	6
16	3	5	5	6	6	7	7	7
17	3	6	6	7	7	8	8	8
18	3	6	7	8	8	9	9	9
19	4	7	8	9	9	10	10	10
20	4	8	9	9	10	10	11	11*
21-25	4	10	11	12	13	13	14*	14*
26-30	4	13	14	16	17	18*	18*	19*
31-35	4	15	18	19	20	21*	22	23
36-40	4	18	20	22	23	24*	25	26
41-45	4	19	22	24	25*	27	28	28
46-50	4	20	23	25	27*	28	29	30
51-55	4	21	24	27	28*	30	31	31
56-60	4	22	25	27	29*	31	32	32
61-65	4	22	25	28	30*	31	32	33
66-70	4	23	26	28	30*	32	33	33

Constants: foliar moisture content = 97% ; crown base height = 18m

Type of fire: **surface** **intermittent crown*** **continuous crown** - CFB 50% **BUI₀**

TABLE 4.6

Equilibrium rate of spread (m/min)
and fire intensity class

C-6 conifer plantation

7 m crown base height

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

BUI

ISI	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
1			1					
2								
3								
4						0.7	0.7	0.7
5					1	1	1	1
6					2	2	2	2
7				2	2	3	3	4
8				3	3	3	6*	7*
9			4	4	4	7*	10*	11*
10			4	5	5	11*	14*	15*
11			5	6	8	15*	18*	19*
12			6	7	13*	19*	21*	22*
13		6	7	8	17*	22*	24*	24*
14		7	8	9	21*	25*	26*	27*
15		8	9	10	24*	28*	29*	29*
16		9	10	16*	27*	30*	31*	31
17		10	11	20*	30*	32*	33	33
18		10	12	24*	32*	34	34	34
19		11	13	28*	34*	35	36	36
20		12	13	30*	35*	37	37	37
21-25		14	16	37*	40	40	40	41
26-30		17	22*	43	45	45	45	45
31-35		19	34*	47	48	48	48	48
36-40		20	41*	50	51	51	51	51
41-45		21	45*	52	53	53	53	53
46-50		22	48*	54	54	54	54	54
51-55		22	50*	55	56	56	56	56
56-60		23	51*	56	57	57	57	57
61-65		23	52*	57	57	57	57	57
66-70		23	53*	58	58	58	58	58

Constants: foliar moisture content = 97%; crown base height = 7m

Type of fire: **surface** **intermittent crown*** **continuous crown** _ CFB 50% **BUI₀**

TABLE 4.7
Equilibrium rate of spread (m/min)
and fire intensity class

C-6 conifer plantation
2 m crown base height

INTENSITY CLASS	
1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

	BUI							
	0	21	31	41	61	81	121	161
ISI	20	30	40	60	80	120	160	200
1		1						
2								
3								
4				2		0.9	1	1
5					2	3*	3*	3*
6				3	4*	5*	6*	6*
7	1			5*	7*	8*	9*	9*
8	1			8*	10*	11*	12*	12*
9	2		6*	11*	13*	14*	15*	15*
10	2		10*	15*	17*	17*	18*	18*
11	3		14*	18*	20*	20*	21*	21*
12	3	6	17*	21*	22*	23*	23*	24*
13	3	11*	21*	24*	25*	25*	26*	26*
14	4	15*	23*	26*	27*	28*	28*	28*
15	4	19*	26*	28*	29*	30	30	30
16	5	22*	28*	30*	31	31	32	32
17	5	25*	31*	32	33	33	33	33
18	5	27*	32*	34	34	35	35	35
19	5	30*	34*	35	36	36	36	36
20	5	32*	36	37	37	37	37	37
21-25	5	37*	40	40	40	41	41	41
26-30	5	43	44	45	45	45	45	45
31-35	5	47	48	48	48	48	48	48
36-40	5	50	51	51	51	51	51	51
41-45	5	52	53	53	53	53	53	53
46-50	5	54	54	54	54	54	54	54
51-55	5	55	56	56	56	56	56	56
56-60	5	56	56	57	57	57	57	57
61-65	5	57	57	57	57	57	57	57
66-70	5	57	58	58	58	58	58	58

Constants: foliar moisture content = 97%; crown base height = 2m

Type of fire: **surface** **intermittent crown*** **continuous crown** _ CFB 50% **[BUI₀]**

TABLE 4.8

Equilibrium rate of spread (m/min)
and fire intensity class

C-7 ponderosa pine-Douglas-fir

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

BUI

ISI	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
1	0	0.1	0.1	0.1	0.1	0.1	0.1	0.1
2	0.1	0.1	0.1	0.1	0.1	0.1	0.2	0.2
3	0.2	0.2	0.2	0.2	0.2	0.2	0.4	0.4
4	0.4	0.4	0.4	0.4	0.4	0.6	0.6	0.6
5	0.4	0.7	0.8	0.8	0.9	1	1	1
6	0.6	1	1	1	1	1	1	1
7	0.8	1	1	2	2	2	2	2
8	1	2	2	2	2	2	2	2
9	1	2	2	2	2	3	3	3
10	2	2	3	3	3	3	3	3
11	2	3	3	3	4	4	4	4
12	2	3	4	4	4	4	4	4
13	2	4	4	4	5	5	5	5
14	3	4	5	5	5	5	6	6
15	3	5	5	6	6	6	6*	6*
16	3	5	6	6	6*	7*	7*	7*
17	4	6	6	7*	7*	7*	8*	8*
18	4	6	7	7*	8*	8*	8*	8*
19	4	7	7*	8*	8*	9*	9*	9*
20	5	7	8*	9*	9*	9*	10*	10*
21-25	6	9*	10*	11*	11*	11*	12*	12*
26-30	7	12*	13*	14*	14*	15*	15*	15*
31-35	9*	14*	16*	17	17	18	18	19
36-40	10*	17*	18	19	20	21	22	22
41-45	12*	19	21	22	23	24	24	25
46-50	13*	21	23	24	26	26	27	27
51-55	14*	23	25	27	28	29	29	30
56-60	15*	24	27	28	30	31	32	32
61-65	16*	26	28	30	32	33	33	34
66-70	17*	27	30	32	33	34	35	36

Constants: foliar moisture content = 97% ; CBH = 10m; surface fuel consumption for FPMC 90

Type of fire: **surface** **intermittent crown*** **continuous crown** _ CFB 50% **BUI₀**

TABLE 4.9
Equilibrium rate of spread (m/min)
and fire intensity class

D-1 leafless aspen

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

	BUI							
	0	21	31	41	61	81	121	161
ISI	20	30	40	60	80	120	160	200
1	1							
2								
3								
4								
5								
6								
7								
8								
9								
10								
11								
12								
13								
14								
15								
16								
17								
18								
19								
20								
21-25								
26-30								
31-35								
36-40								
41-45								
46-50								
51-55								
56-60								
61-65								
66-70								

Note: crown fires are not expected in deciduous fuel types

Type of fire: surface BUI₀

TABLE 4.10

Equilibrium rate of spread (m/min)
and fire intensity class

M-1 boreal mixedwood-leafless
75% conifer / 25% hardwood

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

BUI

ISI	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
1								0.5
2		2		1	1	1	1	1
3		2	2	2	2	2	2	2
4		2	3	3	3	3*	3*	4*
5		3	4	4	4*	5*	5*	5*
6		4	5	5*	6*	6*	6*	6*
7		5	6*	7*	7*	7*	8*	8*
8		6	7*	8*	8*	9*	9*	9*
9	4	7	8*	9*	10*	10*	11*	11*
10	4	9*	10*	11*	11*	12*	12*	13
11	5	10*	11*	12*	13*	13	14	14
12	6	11*	12*	14*	14	15	16	16
13	6	12*	14*	15	16	17	17	18
14	7	13*	15*	16	17	18	19	19
15	7	14*	16	18	19	20	21	21
16	8	15*	18	19	20	21	22	23
17	9	17*	19	21	22	23	24	24
18	9	18	20	22	24	25	25	26
19	10	19	21	24	25	26	27	28
20	10	20	23	25	27	28	29	29
21-25	12	23	26	29	31	32	34	34
26-30	15	29	32	36	38	40	41	42
31-35	18*	34	38	42	44	47	48	49
36-40	20*	38	43	47	50	53	54	55
41-45	22*	42	48	52	56	58	60	61
46-50	24*	46	52	57	60	63	65	67
51-55	26	49	55	61	65	68	70	71
56-60	27	52	59	64	69	72	74	75
61-65	28	54	62	68	72	75	78	79
66-70	30	57	64	70	75	79	81	82

Constants: foliar moisture content = 97%; crown base height = 6m

Type of fire: **surface** **intermittent crown*** **continuous crown** _ CFB 50% **BUI₀**

TABLE 4.11

Equilibrium rate of spread (m/min)
and fire intensity class

M-1 boreal mixedwood - leafless
50% conifer / 50% hardwood

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

	BUI							
	0	21	31	41	61	81	121	161
ISI	20	30	40	60	80	120	160	200
1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1
2	1.1	1.1	1.1	1.1	0.9	0.9	0.9	0.9
3	1.1	1	1	1	2	2	2	2
4	1	2	2	2	2	2	3	3
5	1	2	3	3	3	3	4*	4*
6	1	3	4	4	4	4*	5*	5*
7	1	4	4	5	5*	5*	6*	6*
8	1	5	5	6*	6*	7*	7*	7*
9	1	6	6	7*	7*	8*	8*	8*
10	1	6	7*	8*	8*	9*	9*	9*
11	1	7	8*	9*	10*	10*	10*	11*
12	4	8	9*	10*	11*	11*	12*	12*
13	5	9	10*	11*	12*	12*	13*	13
14	5	10*	11*	12*	13*	14	14	14
15	6	11*	12*	13*	14	15	15	16
16	6	12*	13*	14*	15	16	17	17
17	6	12*	14*	15	16	17	18	18
18	7	13*	15*	17	18	18	19	19
19	7	14*	16*	18	19	20	20	21
20	8	15*	17	19	20	21	22	22
21-25	9	18*	20	22	23	24	25	26
26-30	11	22	24	27	29	30	31	31
31-35	13	25	29	32	34	35	36	37
36-40	15	29	33	36	38	40	41	42
41-45	17	32	36	40	42	44	46	47
46-50	18	35	39	43	46	48	50	51
51-55	19*	37	42	46	49	52	53	54
56-60	21*	40	45	49	52	55	57	58
61-65	22*	42	47	52	55	58	59	60
66-70	23*	43	49	54	57	60	62	63

Constants: foliar moisture content = 97%; crown base height = 6m

Type of fire: **surface** **intermittent crown*** **continuous crown** _ CFB 50% **BUI₀**

TABLE 4.12

Equilibrium rate of spread (m/min)
and fire intensity class

M-1 boreal mixedwood-leafless
25% conifer / 75% hardwood

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

BUI

ISI	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
1	1							
2								
3		2				1	1	1
4				1	2	2	2	2
5			2	2	2	2	2	2
6			2	3	3	3	3	3
7		3	3	3	3	4		4
8		3	3	4	4	4	4*	4*
9		4	4	4	5	5*	5*	5*
10		4	5	5	6	6*	6*	6*
11		5	5	6	6*	7*	7*	7*
12		5	6	7	7*	7*	8*	8*
13		6	7	7*	8*	8*	8*	9*
14		6	7	8*	9*	9*	9*	9*
15		7	8	9*	9*	10*	10*	10*
16		8	9	9*	10*	11*	11*	11*
17		8	9*	10*	11*	11*	12*	12*
18		9	10*	11*	12*	12*	13*	13*
19		9	11*	12*	12*	13*	13*	14
20	5	10	11*	12*	13*	14*	14	15
21-25	6	12*	13*	15*	16	16	17	17
26-30	8	15*	16*	18	19	20	21	21
31-35	9	17*	19	21	23	24	25	25
36-40	10	20*	22	24	26	27	28	28
41-45	11	22	25	27	29	30	31	32
46-50	12	24	27	30	31	33	34	35
51-55	13	26	29	32	34	35	37	37
56-60	14	27	31	34	36	38	39	40
61-65	15	29	32	36	38	40	41	42
66-70	16	30	34	37	40	42	43	44

Constants: foliar moisture content = 97%; crown base height = 6m

Type of fire: surface intermittent crown* continuous crown _ CFB 50% BUI₀

TABLE 4.13

Equilibrium rate of spread (m/min)
and fire intensity class

M-2 boreal mixedwood-green
75% conifer / 25% hardwood

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

ISI	BUI							
	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
2	0.5	0.7	1	1	1	1	1	1
3	1.1	2	2	2	2	2	2	2
4	1	2	3	3	3	3*	3*	3*
5	1	3	4	4	4*	4*	5*	5*
6	2	4	5	5*	5*	6*	6*	6*
7	3	5	6	6*	7*	7*	7*	7*
8	4	6	7*	8*	8*	8*	9*	9*
9	4	7	8*	9*	9*	10*	10*	10*
10	4	8*	9*	10*	11*	11*	12*	12*
11	5	9*	10*	12*	12*	13	13	14
12	5	10*	12*	13*	14	14	15	15
13	6	11*	13*	14*	15	16	16	17
14	7	13*	14*	16	17	17	18	18
15	7	14*	15*	17	18	19	20	20
16	8	15*	17	18	20	20	21	21
17	8	16*	18	20	21	22	23	23
18	9	17*	19	21	22	23	24	25
19	9	18	20	22	24	25	26	26
20	10	19	22	24	25	26	27	28
21-25	12	22	25	28	29	31	32	32
26-30	14	27	31	34	36	38	39	40
31-35	17*	32	36	40	42	44	46	46
36-40	19*	36	41	45	48	50	52	52
41-45	21*	40	45	49	53	55	57	58
46-50	23*	43	49	54	57	60	62	63
51-55	24*	46	52	57	61	64	66	67
56-60	25	49	55	61	65	68	70	71
61-65	27	51	58	64	68	71	73	75
66-70	28	53	60	66	70	74	76	78

Constants: foliar moisture content = 97%; crown base height = 6m

Type of fire: **surface** **intermittent crown*** **continuous crown** — CFB 50% **BUI₀**

TABLE 4.14

Equilibrium rate of spread (m/min)
and fire intensity class**M-2 boreal mixedwood-green**

50% conifer / 50% hardwood

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

BUI

ISI	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
1	4.1	4.2	4.3	4.4	4.5	4.6	4.7	4.8
2	4.3	4.4	4.5	4.6	4.7	4.8	4.9	5.0
3	4.5	4.6	4.7	4.8	4.9	5.0	5.1	5.2
4	4.7	4.8	4.9	5.0	5.1	5.2	5.3	5.4
5	4.9	5.0	5.1	5.2	5.3	5.4	5.5	5.6
6	5.1	5.2	5.3	5.4	5.5	5.6	5.7	5.8
7	5.3	5.4	5.5	5.6	5.7	5.8	5.9	6.0
8	5.5	5.6	5.7	5.8	5.9	6.0	6.1	6.2
9	5.7	5.8	5.9	6.0	6.1	6.2	6.3	6.4
10	5.9	6.0	6.1	6.2	6.3	6.4	6.5	6.6
11	6.1	6.2	6.3	6.4	6.5	6.6	6.7	6.8
12	6.3	6.4	6.5	6.6	6.7	6.8	6.9	7.0
13	6.5	6.6	6.7	6.8	6.9	7.0	7.1	7.2
14	6.7	6.8	6.9	7.0	7.1	7.2	7.3	7.4
15	6.9	7.0	7.1	7.2	7.3	7.4	7.5	7.6
16	7.1	7.2	7.3	7.4	7.5	7.6	7.7	7.8
17	7.3	7.4	7.5	7.6	7.7	7.8	7.9	8.0
18	7.5	7.6	7.7	7.8	7.9	8.0	8.1	8.2
19	7.7	7.8	7.9	8.0	8.1	8.2	8.3	8.4
20	7.9	8.0	8.1	8.2	8.3	8.4	8.5	8.6
21-25	8.1	8.2	8.3	8.4	8.5	8.6	8.7	8.8
26-30	8.3	8.4	8.5	8.6	8.7	8.8	8.9	9.0
31-35	8.5	8.6	8.7	8.8	8.9	9.0	9.1	9.2
36-40	8.7	8.8	8.9	9.0	9.1	9.2	9.3	9.4
41-45	8.9	9.0	9.1	9.2	9.3	9.4	9.5	9.6
46-50	9.1	9.2	9.3	9.4	9.5	9.6	9.7	9.8
51-55	9.3	9.4	9.5	9.6	9.7	9.8	9.9	10.0
56-60	9.5	9.6	9.7	9.8	9.9	10.0	10.1	10.2
61-65	9.7	9.8	9.9	10.0	10.1	10.2	10.3	10.4
66-70	9.9	10.0	10.1	10.2	10.3	10.4	10.5	10.6

Constants: foliar moisture content = 97%; crown base height = 6m

Type of fire: **surface** **intermittent crown*** **continuous crown** - CFB 50% **BU10**

TABLE 4.15

Equilibrium rate of spread (m/min)
and fire intensity class

M-2 boreal mixedwood-green
25% conifer / 75% hardwood

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

	BUI							
	0	21	31	41	61	81	121	161
ISI	20	30	40	60	80	120	160	200
1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	0.8
4	1	2	2	2	1	1	1	1
5	1	2	2	1	2	2	2	2
6	1	2	2	2	2	2	2	2
7	1	2	2	2	2	3	3	3
8	1	2	3	3	3	3	3	3
9	1	3	3	3	3	4	4	4
10	2	3	3	4	4	4	4	4*
11	2	3	4	4	4	5	5*	5*
12	2	4	4	5	5	5*	5*	6*
13	2	4	5	5	6	6*	6*	6*
14	2	5	5	6	6*	6*	7*	7*
15	2	5	6	6	7*	7*	7*	7*
16	2	5	6	7	7*	8*	8*	8*
17	2	6	7	7*	8*	8*	8*	8*
18	2	6	7	8*	8*	9*	9*	9*
19	2	7	7	8*	9*	9*	9*	10*
20	2	7	8	9*	9*	10*	10*	10*
21-25	2	8	9*	10*	11*	11*	12*	12*
26-30	5	10	11*	13*	13*	14*	14	15
31-35	6	12*	13*	15*	16	16	17	17
36-40	7	13*	15*	17	18	19	19	19
41-45	8	15*	17*	18	20	21	21	22
46-50	8	16*	18*	20	21	22	23	23
51-55	9	17*	20	21	23	24	25	25
56-60	10	18*	21	23	24	25	26	27
61-65	10	19*	22	24	25	27	27	28
66-70	10	20*	23	25	26	28	29	29

Constants: foliar moisture content = 97%; crown base height = 6m

Type of fire: **surface** **intermittent crown*** **continuous crown** _ CFB 50% **BU₁₀**

TABLE 4.16

Equilibrium rate of spread (m/min)
and fire intensity class

M-3 dead balsam fir mixedwood
-leafless / 30% dead fir

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

BUI

ISI	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
1	1	1	1	1	1	1	1	1
2	1	2	2	2	2	2	0.5	0.5
3	1	2	2	0.8	0.8	0.9	0.9	0.9
4	1	1	1	1	1	1	1	1
5	1	1	2	2	2	2	2	2
6	1	2	2	2	2	2	3*	3*
7	1	2	3	3	3	3*	3*	3*
8	1	3	3	3	4*	4*	4*	4*
9	1	3	4	4	4*	4*	5*	5*
10	1	4	4	5*	5*	5*	5*	5*
11	1	4	5	5*	6*	6*	6*	6*
12	1	5	5*	6*	6*	7*	7*	7*
13	1	5	6*	7*	7*	7*	8*	8*
14	3	6	7*	7*	8*	8*	8*	9*
15	3	6	7*	8*	9*	9*	9*	9*
16	4	7*	8*	9*	9*	10*	10*	10*
17	4	8*	9*	9*	10*	11*	11*	11*
18	4	8*	9*	10*	11*	11*	12*	12
19	5	9*	10*	11*	12*	12*	13	13
20	5	9*	11*	12*	12*	13	13	14
21-25	6	11*	13*	14	15	15	16	16
26-30	7	14*	16	17	18	19	20	20
31-35	9	17	19	21	22	23	24	24
36-40	10	19	22	24	25	27	28	28
41-45	12	22	24	27	29	30	31	31
46-50	13	24	27	30	31	33	34	35
51-55	14	26	29	32	34	36	37	38
56-60	15*	28	31	34	36	38	39	40
61-65	16*	29	33	36	39	40	42	43
66-70	16*	31	35	38	41	43	44	45

Constants: foliar moisture content = 97%; crown base height = 6m

Type of fire: **surface** **intermittent crown*** **continuous crown** _ CFB 50% **BUI₀**

TABLE 4.17

Equilibrium rate of spread (m/min)
and fire intensity class**M-3** dead balsam fir mixedwood
- leafless / 60% dead fir

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

ISI	BUI							
	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
1	0.4	0.7	0.8	0.8	0.9	1	1	1
2	1	2	2	2	2	3	3*	3*
3	2	3	4	4*	4*	5*	5*	5*
4	3	5	6*	6*	7*	7*	7*	7*
5	4	7*	8*	8*	9*	9*	10*	10*
6	4	9*	10*	11*	11*	12*	12	13
7	5	11*	12*	13*	14	15	15	15
8	7	12*	14*	16	17	17	18	18
9	8	14*	16	18	19	20	21	21
10	9	16	19	20	22	23	24	24
11	10	18	21	23	24	25	26	27
12	11	20	23	25	27	28	29	30
13	12	22	25	28	29	31	32	32
14	13	24	27	30	32	33	35	35
15	14	26	29	32	34	36	37	38
16	15*	28	32	35	37	39	40	41
17	15*	30	34	37	39	41	42	43
18	16*	31	35	39	41	43	45	46
19	17*	33	37	41	44	46	47	48
20	18*	35	39	43	46	48	50	51
21-25	21*	39	45	49	52	55	56	57
26-30	24	46	52	57	61	64	66	67
31-35	27	52	59	64	69	72	74	76
36-40	30	57	64	70	75	78	81	82
41-45	32	60	68	75	80	84	86	88
46-50	33	64	72	79	84	88	91	93
51-55	34	66	75	82	87	92	95	96
56-60	36	68	77	85	90	95	98	99
61-65	36	70	79	87	92	97	100	102
66-70	37	71	81	88	94	99	102	104

Constants: foliar moisture content = 97%; crown base height = 6m

Type of fire: **surface** **intermittent crown*** **continuous crown** _ CFB 50% **[BUI₀]**

TABLE 4.18

Equilibrium rate of spread (m/min)
and fire intensity class

M-3 dead balsam fir mixedwood
- leafless / 100% dead fir

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

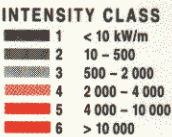
	BUI							
	0	21	31	41	61	81	121	161
ISI	20	30	40	60	80	120	160	200
1	0.9 2	2 3	2	2	2	2	2*	2*
2	2	4	5 4	5*	6*	6*	6*	6*
3	4	7*	8*	9* 5	10*	10*	11*	11*
4	6	11*	12*	13*	14	15	15	15
5	7	14*	16	17	18	19	20	20
6	9	17	20	21	23	24	25	25
7	11	21	23	26	27	29	29	30
8	12	24	27	30	32	33	34	35
9	14*	27	31	34	36	38	39	40
10	16*	30	34	38	40	42	43	44
11	17*	33	38	42	44 6	46	48	49
12	19*	36	41	45	48	50	52	53
13	21*	39	44	49	52	54	56	57
14	22*	42	48	52	56	58	60	61
15	23	45	51	56	59	62	64	65
16	25	47	54	59	63	66	68	69
17	26	50	56	62	66	69	71	73
18	27	52	59	65	69	72	75	76
19	28	54	62	68	72	75	78	79
20	30	57	64	70	75	78	81	82
21-25	33	63	71	78	83	87	89	91
26-30	37	71	80	88	93	98	101	103
31-35	40	77	87	95	102	107	110	112
36-40	43	82	92	101	108	113	117	119
41-45	45	85	97	106	113	118	122	124
46-50	46	88	100	109	117	122	126	128
51-55	47	90	102	112	119	125	129	131
56-60	48	92	104	114	121	127	131	134
61-65	49	93	105	116	123	129	133	135
66-70	49	94	106	117	124	130	134	137

Constants: crown base height = 6m

Type of fire: **surface** **intermittent crown*** **continuous crown** _ CFB 50% **BUI₀**

TABLE 4.19
Equilibrium rate of spread (m/min)
and fire intensity class

M-4 dead balsam fir mixedwood
- green / 30% dead fir



ISI	BUI							
	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
1	1	1	1	1	1	1	1	1
2	1	1	1	1	1	1	1	1
3	1	1	1	1	1	1	1	1
4	1	1	1	1	1	0.5	0.5	0.5
5	1	1	1	1	0.8	0.8	0.8	0.9
6	1	1	1	1	1	1	1	1
7	1	1	1	2	2	2	2	2
8	1	2	2	2	2	2	2	2*
9	1	2	2	3	3	3*	3*	3*
10	1	3	3	3	3*	3*	4*	4*
11	1	3	3	4	4*	4*	4*	4*
12	1	4	4	4*	5*	5*	5*	5*
13	1	4	5	5*	6*	6*	6*	6*
14	1	5	5*	6*	6*	7*	7*	7*
15	1	5	6*	7*	7*	7*	8*	8*
16	3	6	7*	8*	8*	8*	9*	9*
17	4	7*	8*	8*	9*	9*	10*	10*
18	4	7*	8*	9*	10*	10*	11*	11*
19	4	8*	9*	10*	11*	11*	12*	12*
20	5	9*	10*	11*	12*	12*	13	13
21-25	6	11*	12*	14*	14	15	16	16
26-30	7	14*	16	18	19	20	20	21
31-35	9	18	20	22	23	24	25	26
36-40	11	20	23	25	27	28	29	30
41-45	12	23	26	29	31	32	33	34
46-50	13	25	29	31	33	35	36	37
51-55	14*	27	31	34	36	38	39	40
56-60	15*	29	33	36	38	40	41	42
61-65	16*	30	34	38	40	42	43	44
66-70	16*	31	36	39	42	44	45	46

Constants: foliar moisture content = 97%; crown base height = 6m
 Type of fire: **surface** **intermittent crown*** **continuous crown** _ CFB 50% **BU₀**

TABLE 4.20

Equilibrium rate of spread (m/min)
and fire intensity class

M-4 dead balsam fir mixedwood
- green / 60% dead fir

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

BUI

ISI	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
1	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
2	0.2	0.2	0.2	0.2	0.2	0.5	0.6	0.6
3	0.3	0.3	1	1	1	1	1	1
4	0.4	2	2	2	2	2	2	2
5	0.5	2	3	3	3	3*	3*	3*
6	0.6	3	4	4	4*	4*	5*	5*
7	0.7	4	5	5*	5*	6*	6*	6*
8	0.8	5	6*	6*	7*	7*	7*	7*
9	3	6	7*	8*	8*	9*	9*	9*
10	4	7*	8*	9*	10*	10*	11*	11*
11	4	9*	10*	11*	11*	12*	12	13
12	5	10*	11*	12*	13	14	14	14
13	6	11*	13*	14	15	15	16	16
14	6	12*	14*	15	16	17	18	18
15	7	14*	15	17	18	19	20	20
16	8	15*	17	19	20	21	21	22
17	9	16	18	20	22	23	23	24
18	9	18	20	22	23	24	25	26
19	10	19	21	24	25	26	27	28
20	11	20	23	25	27	28	29	29
21-25	13	24	27	30	32	33	34	35
26-30	16*	30	34	37	40	42	43	44
31-35	18*	35	40	44	47	49	51	52
36-40	21*	40	45	50	53	56	57	58
41-45	23*	44	50	55	58	61	63	64
46-50	25	48	54	59	63	66	68	69
51-55	26	51	57	63	67	70	72	74
56-60	28	53	60	66	70	73	76	77
61-65	29	55	62	68	73	76	79	80
66-70	30	57	64	70	75	79	81	83

Constants: foliar moisture content = 97%; crown base height = 6m

Type of fire: **surface** **intermittent crown*** **continuous crown** _ CFB 50% **BUI₀**

TABLE 4.21

Equilibrium rate of spread (m/min)
and fire intensity class

M-4 dead balsam fir mixedwood
- green /100% dead fir

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

ISI	BUI							
	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
1	0.4	0.7	0.8	0.8	0.9	0.9	1	1
2	1	2	2	2	2	3	3*	3*
3	2	3	4	4	4*	5*	5*	5*
4	3	5	5*	6*	6*	7*	7*	7*
5	3	7	7*	8*	9*	9*	9*	10*
6	4	8*	9*	10*	11*	12*	12*	12
7	5	10*	12*	13*	13	14	15	15
8	6	12*	14*	15	16	17	17	18
9	7	14*	16	17	18	19	20	20
10	8	16*	18	20	21	22	23	23
11	9	18	20	22	23	25	25	26
12	10	20	22	24	26	27	28	29
13	11	22	24	27	28	30	31	31
14	12	23	26	29	31	32	33	34
15	13	25	28	31	33	35	36	37
16	14*	27	31	33	36	37	39	39
17	15*	29	32	36	38	40	41	42
18	16*	30	34	38	40	42	44	44
19	17*	32	36	40	42	45	46	47
20	18*	34	38	42	45	47	48	49
21-25	20*	38	43	48	51	53	55	56
26-30	24*	45	51	56	60	63	65	65
31-35	27	51	58	64	68	71	74	75
36-40	29	56	64	70	75	78	81	82
41-45	32	61	68	75	80	84	87	88
46-50	34	64	73	80	85	89	92	94
51-55	35	67	76	83	89	93	96	98
56-60	36	70	79	86	92	96	100	101
61-65	37	72	81	89	95	100	102	104
66-70	38	73	83	91	97	105	105	106

Constants: crown base height = 6m

Type of fire: **surface** **intermittent crown*** **continuous crown** _ CFB 50% **BUI₀**

TABLE 4.22

Equilibrium rate of spread (m/min)
and fire intensity class**S-1 jack or lodgepole pine slash**

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

BUI

	0	21	31	41	61	81	121	161
ISI	20	30	40	60	80	120	160	200
1	0.3	0.6	0.7	0.8	0.9	1	1	1
2	0.7	2	2	2	2	2	2	2
3	1	3	3	3	4	4	4	4
4	2	4	4	5	5	6	6	6
5	2	5	6	6	7	7	8	8
6	3	6	7	8	8	9	9	10
7	3	7	8	9	10	11	11	11
8	4	8	10	11	12	13	13	13
9	4	9	11	13	14	14	15	15
10	5	11	12	14	15	16	17	17
11	5	12	14	16	17	18	19	19
12	6	13	15	17	19	20	21	21
13	6	14	17	19	20	22	23	23
14	7	15	18	20	22	23	24	25
15	7	16	19	22	24	25	26	27
16	8	18	21	23	25	27	28	29
17	8	19	22	25	27	29	30	30
18	9	20	23	26	28	30	31	32
19	9	21	24	28	30	32	33	34
20	9	22	26	29	31	33	35	36
21-25	11	25	29	33	36	38	40	41
26-30	13	30	35	39	42	45	47	48
31-35	15	34	40	45	48	51	54	55
36-40	16	37	44	50	54	57	59	61
41-45	18	41	48	54	58	62	65	66
46-50	19	44	51	58	62	66	69	71
51-55	20	46	54	61	66	70	73	75
56-60	21	48	57	64	69	73	77	78
61-65	22	50	59	66	72	76	80	81
66-70	22	52	61	68	74	79	82	84

Type of fire: surface BUI₀

TABLE 4.23

Equilibrium rate of spread (m/min)
and fire intensity class

S-2 white spruce-balsam slash

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

ISI	BUI							
	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
1	0.1	0.1	0.2	0.2	0.2	0.2	0.2	0.2
2	0.2	0.4	0.5	0.6	0.6	0.6	0.7	0.7
3	0.4	0.8	1	1	1	1	1	1
4	0.6	1	1	2	2	2	2	2
5	0.8	2	2	2	3	3	3	3
6	1	2	3	3	3	4	4	4
7	1	3	3	4	4	5	5	5
8	2	4	4	5	5	5	6	6
9	2	4	5	6	6	6	7	7
10	2	5	6	6	7	7	8	8
11	2	6	7	7	8	8	9	9
12	3	6	7	8	9	10	10	10
13	3	7	8	9	10	11	11	11
14	3	8	9	10	11	12	12	12
15	4	8	10	11	12	13	13	13
16	4	9	10	12	13	14	14	14
17	4	10	11	13	14	15	15	16
18	4	10	12	13	15	16	16	17
19	5	11	13	14	16	16	17	18
20	5	11	13	15	16	17	18	19
21-25	6	13	15	17	19	20	21	21
26-30	7	16	19	21	23	24	25	26
31-35	8	18	21	24	26	28	29	29
36-40	9	20	23	26	29	30	32	32
41-45	9	22	25	29	31	33	34	35
46-50	10	23	27	30	33	35	36	37
51-55	10	24	28	32	34	37	38	39
56-60	11	25	29	33	36	38	39	40
61-65	11	26	30	34	37	39	41	42
66-70	11	26	31	35	38	40	42	42

Type of fire: surface BUI₀

TABLE 4.24

Equilibrium rate of spread (m/min)
and fire intensity class**S-3** coastal cedar-hemlock-
Douglas-fir slash

INTENSITY CLASS

1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

BUI

ISI	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
1	0	0	0	0	0	0	0	0
2	0.1	0.1	0.1	0.2	0.2	0.2	0.2	0.2
3	0.2	0.4	0.5	0.5	0.6	0.6	0.6	0.6
4	0.4	0.9	1	1	1	1	1	1
5	0.7	2	2	2	2	2	2	3
6	1	2	3	3	4	4	4	4
7	2	4	4	5	5	5	6	6
8	2	5	6	6	7	7	8	8
9	3	6	7	8	9	10	10	10
10	3	8	9	10	11	12	13	13
11	4	10	11	13	14	15	15	16
12	5	11	13	15	16	17	18	18
13	6	13	15	17	19	20	21	21
14	6	15	18	20	21	23	24	24
15	7	17	20	22	24	26	27	27
16	8	19	22	25	27	28	29	30
17	9	20	24	27	29	31	32	33
18	10	22	26	29	32	34	35	36
19	10	24	28	31	34	36	38	38
20	11	25	30	33	36	39	40	41
21-25	13	30	35	39	43	45	47	48
26-30	15	36	42	47	51	54	57	58
31-35	17	40	47	53	58	61	64	65
36-40	19	43	51	57	62	66	69	70
41-45	20	45	53	60	65	69	72	74
46-50	20	47	55	62	67	71	74	76
51-55	21	48	56	63	69	73	76	78
56-60	21	48	57	64	69	74	77	79
61-65	21	49	57	65	70	75	78	79
66-70	21	49	58	65	71	75	78	80

Type of fire: surface **BUI₀**

TABLE 4.25
Equilibrium rate of spread (m/min)
and fire intensity class
O-1a matted grass

INTENSITY CLASS	
1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

ISI	Degree of curing (%)					
	0 50	51 60	61 70	71 80	81 90	91 100
1		1.3	2.4	3.7	5	6
2		1.4	2	3	5	6
3		1.7	2	3	5	6
4	1	1	2	3	7	9
5		1	2	6	9	11
6		2	3	8	11	14
7		2	6	10	14	18
8		3	7	12	16	21
9		3	8	13	19	24
10		3	9	15	21	27
11		3	10	17	24	30
12		3	11	19	26	34
13		4	13	21	29	37
14		4	14	22	31	40
15		5	15	24	34	43
16		6	16	26	36	46
17		6	17	28	39	50
18		6	18	30	41	53
19		7	19	31	43	56
20		7	20	33	46	59
21-25		8	23	38	53	67
26-30		10	27	45	63	81
31-35		11	32	52	72	93
36-40		12	35	58	81	103
41-45		14	38	63	88	113
46-50		15	41	68	94	121
51-55		15	44	72	100	128
56-60		16	46	75	105	134
61-65		17	48	78	109	140
66-70		17	49	81	113	144

Constants: surface fuel load = 3t/ha
 Type of fire: **surface**

TABLE 4.26
Equilibrium rate of spread (m/min)
and fire intensity class
O-1b standing grass

INTENSITY CLASS	
1	< 10 kW/m
2	10 - 500
3	500 - 2 000
4	2 000 - 4 000
5	4 000 - 10 000
6	> 10 000

		Degree of curing (%)					
		0	51	61	71	81	91
ISI		50	60	70	80	90	100
1							
2		1					
3							
4			2			6	7
5					6	8	10
6					8	10	13
7				6	10	13	17
8				7	12	16	21
9				8	14	19	25
10				10	16	22	29
11				11	18	26	33
12				13	21	29	37
13				14	23	32	41
14				15	25	35	45
15			6	17	28	39	50
16			7	18	30	42	54
17			7	20	33	45	58
18			8	21	35	49	62
19			8	23	37	52	67
20			9	24	40	55	71
21-25			10	28	47	65	83
26-30			12	35	57	80	102
31-35			14	41	67	93	120
36-40			16	46	76	105	135
41-45			18	51	83	116	148
46-50			19	55	90	125	160
51-55			21	58	95	133	170
56-60			22	61	100	140	179
61-65			23	64	105	146	187
66-70			23	66	108	150	193

Constants: surface fuel load = 3t/ha

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TABLE 5.1

Spread distance (m)

equilibrium ROS: all fuel types
accelerating ROS: open fuels and
surface fires in closed fuel types

equil ROS	Elapsed time											
	15 min		30 min		45 min		1h		2h		3h	
0.2	3	2	6	4	9	7	12	10	24	22	36	34
0.4	6	3	12	9	18	15	24	21	48	45	72	69
0.6	9	5	18	13	27	22	36	31	72	87	108	103
0.8	12	6	24	17	36	29	48	41	96	89	144	137
1	15	8	30	22	45	36	60	51	120	111	180	171
2	30	16	60	43	90	73	120	103	240	223	360	343
3	45	24	90	65	135	109	180	154	360	334	540	514
4	60	31	120	86	180	145	240	205	480	445	720	685
5	75	39	150	108	225	182	300	257	600	557	900	857
6	90	47	180	129	270	218	360	308	720	668	1080	1028
7	105	55	210	151	315	254	420	359	840	779	1260	1199
8	120	63	240	173	360	291	480	411	960	890	1440	1370
9	135	71	270	194	405	327	540	462	1080	1002	1620	1542
10	150	79	300	216	450	364	600	513	1200	1113	1800	1713
12	180	94	360	259	540	436	720	616	1440	1336	2160	2056
14	210	110	420	302	630	509	840	718	1680	1558	2520	2398
16	240	126	480	345	720	582	960	821	1920	1781	2880	2741
18	270	141	540	388	810	654	1080	924	2160	2003	3240	3083
20	300	157	600	432	900	727	1200	1026	2400	2226	3600	3426
25	375	196	750	540	1125	909	1500	1263	3000	2783	4500	4283
30	450	236	900	647	1350	1091	1800	1539	3600	3339	5400	5139
35	525	275	1050	755	1575	1272	2100	1796	4200	3896	6300	5996
40	600	314	1200	863	1800	1454	2400	2053	4800	4452	7200	6852
45	675	353	1350	971	2025	1636	2700	2309	5400	5009	8100	7709
50	750	393	1500	1079	2250	1818	3000	2566	6000	5565	9000	8565
55	825	432	1650	1187	2475	1999	3300	2822	6600	6122	9900	9422
60	900	471	1800	1295	2700	2181	3600	3079	7200	6678	10800	10278
65	975	510	1950	1403	2925	2363	3900	3335	7800	7235	11700	11135
70	1050	550	2100	1511	3150	2545	4200	3592	8400	7791	12600	11991
75	1125	589	2250	1619	3375	2727	4500	3848	9000	8348	13500	12848
80	1200	628	2400	1726	3600	2908	4800	4105	9600	8904	14400	13704
85	1275	668	2550	1834	3825	3090	5100	4362	10200	9461	15300	14561
90	1350	707	2700	1942	4050	3272	5400	4618	10800	10017	16200	15417
95	1425	746	2850	2042	4275	3454	5700	4875	11400	10574	17100	16417
100	1500	785	3000	2168	4500	3635	6000	5131	12000	11130	18000	17130

Red figures are distances for equilibrium ROS in all fuel types

Black figures are distances for accelerating ROS in open fuel types and surface fires in closed fuel types. Open fuel types are C-1, D-1, O-1, S-1, S-2, S-3 (C-2, C-7). C-2 and C-7 are considered open types when the crown closure is less than 50%.

TABLE 5.2

Spread distance_(m)

accelerating ROS:
crown fires in closed fuel types

equil. ROS	Elapsed time											
	15 min		30 min		45 min		1 h		2 h		3 h	
0.2	1	1	3	4	6	7	8	10	21	22	32	34
0.4	2	3	8	8	11	14	17	20	41	44	65	68
0.6	3	4	9	12	17	21	25	30	61	66	97	102
0.8	4	6	12	17	23	28	34	40	81	88	129	136
1	5	7	15	21	28	35	42	50	102	110	162	170
2	9	15	30	42	56	71	84	101	203	221	323	341
3	14	22	46	62	84	106	127	151	305	331	485	511
4	19	30	61	83	113	142	169	202	406	442	646	682
5	24	37	76	104	141	177	211	252	508	552	808	852
6	28	44	91	125	169	213	253	303	609	662	969	1022
7	33	52	106	146	197	248	296	353	711	773	1131	1193
8	38	59	121	167	225	284	338	403	812	883	1292	1363
9	43	67	137	187	253	319	380	454	914	994	1454	1534
10	47	74	152	208	281	355	422	504	1016	1104	1615	1704
12	57	89	182	250	338	426	507	605	1219	1325	1938	2045
14	66	104	212	292	394	497	591	706	1422	1546	2261	2386
16	78	119	243	333	450	568	676	807	1625	1766	2584	2726
18	85	133	273	375	507	639	780	936	1828	1987	2907	3067
20	95	148	303	417	563	710	845	1008	2031	2208	3231	3408
25	118	185	379	521	704	887	1068	1281	2539	2760	4038	4260
30	142	222	455	625	884	1065	1267	1513	3047	3312	4846	5112
35	165	260	531	729	985	1242	1479	1785	3554	3864	5653	5964
40	189	297	607	833	1126	1420	1690	2017	4062	4416	6461	6816
45	213	334	683	937	1266	1597	1901	2269	4570	4968	7269	7668
50	236	371	758	1041	1407	1775	2112	2521	5078	5520	8076	8520
55	260	406	834	1145	1548	1952	2323	2773	5585	6072	8884	9372
60	284	445	910	1250	1689	2130	2535	3025	6093	6624	9692	10224
65	307	482	986	1354	1829	2307	2746	3277	6601	7176	10499	11076
70	331	519	1062	1458	1970	2484	2957	3530	7109	7728	11307	11928
75	355	556	1138	1562	2111	2662	3168	3782	7617	8280	12115	12780
80	378	593	1213	1668	2251	2839	3380	4034	8124	8832	12922	13632
85	402	630	1288	1770	2392	3017	3591	4286	8632	9384	13730	14484
90	426	667	1365	1874	2533	3194	3802	4538	9140	9936	14537	15336
95	449	704	1441	1978	2674	3372	4013	4790	9648	10488	15345	16188
100	473	741	1517	2083	2814	3549	4224	5042	10155	11040	16153	17040

Black figures are distances for intermittent crown fires with accelerating ROS (50% CFB)

Red figures are distances for continuous crown fires with accelerating ROS (90% CFB)

TABLE 6.1

Fire Area(ha)

Perimeter(m)

timber and slash fuel types

Spread

Effective wind speed (km/h)

Dist.	0	5	10	15	20	25	30	35	40	45	50
25	<0.1 79	<0.1 74	<0.1 66	<0.1 61	<0.1 57	<0.1 55	<0.1 54	<0.1 53	<0.1 52	<0.1 52	<0.1 52
50	0.2 157	0.2 148	0.1 133	0.1 121	0.1 114	0.1 110	<0.1 108	<0.1 106	<0.1 105	<0.1 104	<0.1 103
100	1 314	1 297	1 266	0.4 243	0.3 229	0.2 220	0.2 215	0.2 212	0.2 209	0.1 208	0.1 206
150	2 471	2 445	1 399	1 364	1 343	1 330	0.5 323	0.4 317	0.4 314	0.3 312	0.3 310
200	3 628	3 594	2 532	2 486	1 457	1 441	1 430	1 423	1 419	1 415	1 413
250	5 785	4 742	3 665	2 607	2 572	2 551	1 538	1 529	1 523	1 519	1 516
300	7 942	6 891	5 798	4 728	3 686	2 661	2 645	2 635	1 628	1 623	1 619
400	13 1257	11 1188	9 1064	6 971	5 915	4 881	3 860	3 847	2 837	2 831	2 826
500	20 1571	17 1485	13 1330	10 1214	8 1144	6 1102	5 1075	4 1058	4 1047	4 1038	3 1032
600	28 1885	25 1782	19 1596	14 1457	11 1372	9 1322	7 1290	6 1270	6 1256	5 1246	5 1239
800	50 2513	45 2376	34 2128	25 1943	20 1830	16 1762	13 1720	11 1693	10 1675	9 1661	8 1652
1000	79 3142	70 2970	53 2660	40 2428	31 2287	25 2203	20 2151	18 2116	16 2093	14 2077	13 2065
1200	113 3770	101 3564	77 3192	57 2914	44 2745	35 2644	29 2581	25 2540	22 2512	20 2492	19 2478
1500	177 4712	157 4455	120 3989	89 3642	69 3431	55 3305	46 3226	40 3175	35 3140	32 3115	29 3097
2000	314 6283	279 5840	213 5319	159 4857	122 4575	98 4406	82 4301	71 4233	62 4187	56 4154	52 4130
2500	491 7854	436 7425	333 6649	248 6071	191 5718	153 5508	128 5376	110 5291	98 5233	88 5192	81 5162
3000	707 9425	628 8910	479 7979	357 7285	275 6882	221 6609	184 6452	159 6349	140 6280	127 6231	117 6195
4000	1257 12566	1117 11880	852 10638	635 9713	489 9149	393 8812	328 8602	282 8466	250 8373	226 8307	207 8260
5000	1983 15708	1748 14849	1331 13298	992 12141	764 11437	613 11015	512 10753	441 10582	390 10466	352 10384	324 10324
L/B:	1.0	1.1	1.5	2.0	2.6	3.3	3.8	4.4	5.0	5.6	6.1

Red figures are area

Black figures are perimeter length

TABLE 6.2

Fire Area (ha)

Perimeter (m) grass fuel types

Spread

Effective wind speed (km/h)

Dist.	0	5	10	15	20	25	30	35	40	45	50
25	<0.1 79	<0.1 58	<0.1 55	<0.1 54	<0.1 53	<0.1 52	<0.1 52	<0.1 52	<0.1 52	<0.1 51	<0.1 51
50	0.2 157	0.1 117	0.1 110	<0.1 107	<0.1 106	<0.1 105	<0.1 104	<0.1 104	<0.1 103	<0.1 103	<0.1 103
100	1 314	0.3 234	0.2 220	0.2 215	0.2 212	0.2 210	0.1 208	0.1 207	0.1 206	0.1 206	0.1 205.2
150	2 471	1 350	1 330	0.5 322	0.4 318	0.4 315	0.3 313	0.3 311	0.3 310	0.3 309	0.3 308
200	3 628	1 467	1 441	1 430	1 424	1 420	1 417	1 415	1 413	0.5 411	0.5 410
250	5 785	2 584	2 551	1 537	1 530	1 525	1 521	1 518	1 516	1 514	1 513
300	7 942	3 701	2 661	2 645	2 635	1 629	1 625	1 622	1 619	1 617	1 615
400	13 1257	5 935	4 881	3 859	3 847	3 839	2 833	2 829	2 826	2 823	2 821
500	20 1571	8 1168	6 1101	5 1074	4 1059	4 1049	4 1042	3 1036	3 1032	3 1029	3 1026
600	28 1885	12 1402	9 1322	7 1289	6 1271	6 1259	5 1250	5 1244	5 1239	4 1234	4 1231
800	50 2513	22 1869	16 1762	13 1719	11 1694	10 1678	9 1667	9 1658	8 1651	8 1646	7 1641
1000	79 3142	34 2336	25 2203	20 2149	18 2118	16 2098	15 2084	14 2073	13 2064	12 2057	12 2052
1200	113 3770	49 2804	35 2643	29 2578	26 2542	23 2518	21 2500	20 2487	18 2477	18 2469	17 2462
1500	177 4712	76 3505	55 3304	46 3223	40 3177	36 3147	33 3126	31 3109	29 3096	27 3086	26 3077
2000	314 6283	135 4673	98 4406	81 4287	71 4236	64 4196	59 4167	55 4146	52 4129	49 4115	46 4103
2500	491 7854	211 5841	153 5507	127 5372	111 5295	100 5245	92 5209	86 5182	81 5161	76 5143	73 5129
3000	707 9425	305 7009	221 6609	183 6446	160 6354	144 6294	133 6251	123 6219	116 6193	110 6172	105 6155
4000	1257 12566	541 9346	392 8812	325 8595	285 8472	257 8392	236 8335	219 8291	206 8257	195 8229	186 8206
5000	1963 15708	846 11682	613 11015	508 10743	445 10590	401 10490	368 10419	343 10364	322 10322	305 10287	291 10258
L/B:	1.0	2.3	3.2	3.5	4.4	4.9	5.3	5.7	6.1	6.4	6.8

Red figures are area

Black figures are perimeter length

TABLE 7.1

Perimeter growth rate (m/min)

PGR timber and slash fuel types

equil.	Effective wind speed (km/h)										
ROS	0	5	10	15	20	25	30	35	40	45	50
0.2	1	1	1	<1	<1	<1	<1	<1	<1	<1	<1
0.4	1	1	1	1	1	1	1	1	1	1	1
0.6	2	2	2	1	1	1	1	1	1	1	1
0.8	3	2	2	2	2	2	2	2	2	2	2
1	3	3	3	2	2	2	2	2	2	2	2
2	6	6	5	5	5	4	4	4	4	4	4
3	9	9	8	7	7	7	6	6	6	6	6
4	13	12	11	10	9	9	9	8	8	8	8
5	16	15	13	12	11	11	11	11	10	10	10
6	19	18	16	15	14	13	13	13	13	12	12
7	22	21	19	17	16	15	15	15	15	15	14
8	25	24	21	19	18	18	17	17	17	17	17
9	28	27	24	22	21	20	19	19	19	19	19
10	31	30	27	24	23	22	22	21	21	21	21
12	38	36	32	29	27	26	26	25	25	25	25
13	41	39	35	32	30	29	28	28	27	27	27
14	44	42	37	34	32	31	30	30	29	29	29
15	47	45	40	36	34	33	32	32	31	31	31
16	50	48	43	39	37	35	34	34	33	33	33
17	53	50	45	41	39	37	37	36	36	35	35
18	57	53	48	44	41	40	39	38	38	37	37
20	63	59	53	49	46	44	43	42	42	42	41
25	79	74	66	61	57	55	54	53	52	52	52
30	94	89	80	73	69	66	65	63	63	62	62
35	110	104	93	85	80	77	75	74	73	73	72
40	126	119	106	97	91	88	86	85	84	83	83
45	141	134	120	109	103	99	97	95	94	93	43
50	157	148	133	121	114	110	108	106	105	104	103
55	173	163	146	134	126	121	118	116	115	114	114
60	188	178	160	146	137	132	129	127	126	125	124
65	204	193	173	158	149	143	140	138	136	135	134
70	220	208	186	170	160	154	151	148	147	145	145
75	236	223	199	182	172	165	161	159	157	156	155
80	251	238	213	194	183	176	172	169	167	166	165
85	267	252	239	219	206	198	194	190	188	187	186
90	283	267	239	219	206	198	194	190	188	187	186
95	298	282	253	231	217	209	204	201	199	197	196
100	314	297	266	243	229	220	215	212	209	208	206

TABLE 7.2

Perimeter growth rate (m/min)

PGR grass fuel types

equil.	Effective wind speed (km/h)										
ROS	0	5	10	15	20	25	30	35	40	45	50
0.2	1	<1	<1	<1	<1	<1	<1	<1	<1	<1	<1
0.4	1	1	1	1	1	1	1	1	1	1	1
0.6	2	1	1	1	1	1	1	1	1	1	1
0.8	3	2	2	2	2	2	2	2	2	2	2
1	3	2	2	2	2	2	2	2	2	2	2
2	6	5	4	4	4	4	4	4	4	4	4
3	9	7	7	6	6	6	6	6	6	6	6
4	13	9	9	9	8	8	8	8	8	8	8
5	16	12	11	11	11	10	10	10	10	10	10
6	19	14	13	13	13	13	13	12	12	12	12
7	22	16	15	15	15	15	15	15	14	14	14
8	25	19	18	17	17	17	17	17	17	16	16
9	28	21	20	19	19	19	19	19	19	19	18
10	31	23	22	21	21	21	21	21	21	21	21
12	38	28	26	26	25	25	25	25	25	25	25
13	41	30	29	28	28	27	27	27	27	27	27
14	44	33	31	30	30	29	29	29	29	29	29
15	47	35	33	32	32	31	31	31	31	31	31
16	50	37	35	34	34	34	33	33	33	33	33
17	53	40	37	37	36	36	35	35	35	35	35
18	57	42	40	39	38	38	38	37	37	37	37
20	63	47	44	43	42	42	42	41	41	41	41
25	79	58	55	54	53	52	52	52	52	51	51
30	94	70	66	64	64	63	63	62	62	62	62
35	110	82	77	75	74	73	73	73	72	72	72
40	126	93	88	86	85	84	83	83	83	82	82
45	141	105	99	97	95	94	94	93	93	93	92
50	157	117	110	107	106	105	104	104	103	103	103
55	173	129	121	118	116	115	115	114	114	113	113
60	188	140	132	129	127	126	125	124	124	123	123
65	204	152	143	140	138	136	135	135	134	134	133
70	220	164	154	150	148	147	146	145	145	144	144
75	236	175	165	161	159	157	156	155	155	154	154
80	251	187	176	172	169	168	167	166	165	165	164
85	267	199	187	183	180	178	177	176	175	175	174
90	283	210	198	193	191	189	188	187	186	185	185
95	298	222	209	204	201	199	198	197	196	195	195
100	314	234	220	215	212	210	208	207	206	206	205

References

- Alexander, M.E. and F.V. Cole. 1995. Head fire intensity class graph for FBP System fuel type C-2 (Boreal Spruce). Can. For. Serv. North. For. Cent., Edmonton, Alta. and Alaska Dep. Nat. Resources, Div. of For., Fairbanks. Poster (with text).
- Alexander, M.E. and W.J. De Groot. 1988. Fire behavior in jack pine stands as related to the Canadian Forest Fire Weather Index System. Can. For. Serv. North. For. Cent., Edmonton, Alta. Poster (with text).
- Alexander, M.E. and R.A. Lanoville. 1989. Predicting fire behavior in the black spruce-lichen woodland fuel type of western and northern Canada. For. Can., North. For. Cent., Edmonton, Alta., and Gov. N.W.T., Dep. Renewable Resour., Territorial For. Fire Cent., Fort Smith, N.W.T. Poster (with text).
- Alexander, M.E., B.D. Lawson, B.J. Stocks, C.E. Van Wagner. 1984. User guide to the Canadian Forest Fire Behavior Prediction System: Rate of spread relationships. Interim edition. Environ. Can., Can. For. Serv., Fire Danger Group, Ottawa. 73 p.
- De Groot, W.J. 1993. Examples of fuel types in the Canadian Forest Fire Behavior Prediction System. For. Can., Nor. For. Cent., Edmonton, Alta. poster (with text).
- Forestry Canada Fire Danger Group. 1992. Development and structure of the Canadian Forest Fire Behavior Prediction System. For. Can., Inf. Rep. ST-X-3. Ottawa 64 p.
- Hirsch, K. G. 1996. Canadian Forest Fire Behavior Prediction (FBP) System User Guide. Nat. Res. Can., Can. For. Serv., Nor. For. Cent. Special Rep. Edmonton, Alta. In Review.
- Lawson, B.D., O.B. Armitage and W. Hoskins, 1996. Diurnal variation in the Fine Fuel Moisture Code: Tables and source code. Canada - B.C. For. Res. Dev. Agreement Report. Victoria, B.C. In Press.
- List, R.J. 1951. Smithsonian meteorological tables. 6th rev. ed. Smithsonian Institution Press, Washington, D.C. 527 p.
- Merrill, D.F. and M.E. Alexander, eds. 1987. Glossary of forest fire management terms. 4th ed. Natl. Res. Counc. Can., Can. Comm. on For. Fire Mgmt., Ottawa. Publ. NRCC 265216. 91 p.
- McAlpine, R.S. 1986. Forest fire growth calculator. Can. For. Serv., North. For. Cent., Edmonton, Alta. For. Manage. Note 43. 3p.

Appendix 1 Symbols

BISI	Back fire Initial Spread Index
BUI	Buildup Index
BROS	Back fire rate of spread (m/min)
CFB	Crown fraction burned
CBH	Crown base height
FFMC	Fine Fuel Moisture Code
ISI	Initial Spread Index
L/B	Length/breadth ratio
PC	Percent conifer
PDF	Percent dead fir
PH	Percent hardwood
PGR	Perimeter growth rate (m/min)
ROS	Head fire rate of spread (m/min)

Appendix 2 Glossary

Buildup Index - A numerical rating of the total amount of fuel available for combustion that combines DMC and DC. In the FBP System, a buildup effect is applied to the primary ISI-based spread equations to represent the effect of increasing fuel consumption on rate of spread. BUI_0 is the BUI value where this effect is zero and rate of spread depends wholly on the Initial Spread Index.

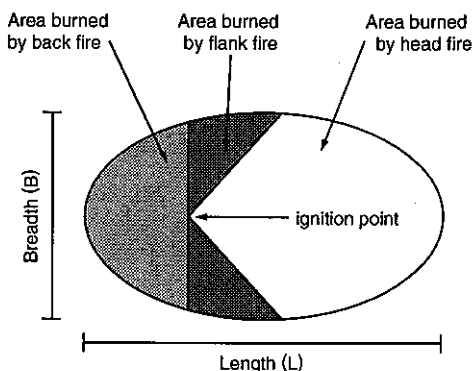
Crown fraction burned - The crown fraction burned is the proportion of tree crowns involved in the fire in a given area. The following descriptive classes are recognized in the FBP System:

Crown fraction burned	Type of fire
< 10%	surface fire
10-89%	intermittent crown fire
> 90%	continuous crown fire

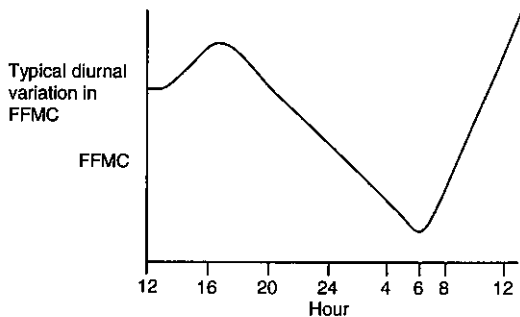
Crown base height - The height, above ground, where the live crown of coniferous trees begins. This value is held constant in each FBP System fuel type except C-6.

live wind speed - The sum of the vectors of the observed wind speed and slope equivalent wind speed.

Elliptical fire growth model - Theory: a free-burning point source fire is an ellipse when fuels are uniform and continuous, topography is homogeneous, windspeed is constant and unidirectional, and the fire is unaffected by suppression activities. The length of the ellipse is the sum of the head fire spread distance and the back fire spread distance. Fire shape or length/breadth ratio (L/B) is determined by the wind speed. Fire area and perimeter length are calculated from the total spread distance and the L/B ratio.



Fine Fuel Moisture Code - A numerical rating of the moisture content of litter and other cured fine fuels. This code indicates the relative ease of ignition and flammability of fine fuel. Fine fuel moisture content varies over a daily or diurnal cycle. The daily standard FFMC is calculated from noon weather observations, but represents fine fuel moisture at 1600 LST, when the fine fuel moisture content is at or near the daily minimum. The diurnal FFMC is an estimate of the FFMC at a particular hour, based on the typical daily variation in temperature and relative humidity in the absence of rain. The hourly FFMC is calculated from actual hourly weather observations.



Fire intensity - The rate of heat energy release per unit time per unit length of fire front. Frontal fire intensity is a major determinant of certain fire effects and difficulty of control. Numerically, it is equal to the product of the net heat of combustion, the quantity of fuel consumed in the flaming front and the linear rate of spread.

Fire perimeter - The entire outer edge or boundary of a fire. See elliptical fire growth model.

Foliar moisture content - The percent moisture content by weight of live conifer needles that are at least one year old.

Initial Spread Index - A numerical rating of the expected rate of fire spread. It combines the effects of wind, slope and FPMC on rate of spread but excludes the influence of the variable quantities of fuel.

Rate of spread - The speed at which a fire extends its horizontal dimensions, expressed in terms of distance per unit of time. Generally thought of in terms of a fire's forward movement, or **head fire rate of spread**, but also applicable to **back fire** and **flank fire ROS**.

Slope equivalent wind speed - An approach used in the FBP System whereby the effect of slope on fire spread with zero wind is given a value in units of wind speed.

Type of fire - See crown fraction burned.

Appendix 3 Examples of FBP System fuel types¹

The Canadian Forest Fire Behavior Prediction (FBP) System models rate of spread, fuel consumption, fire intensity, and fire growth for 16 national benchmark fuel types. A detailed description of the FBP System fuel types (stand structure, surface and ladder fuels) can be found in Information Report ST-X-3, Development and Structure of the Canadian Forest Fire Behavior Prediction System (Forestry Canada Fire Danger Group 1992). The following photos show the benchmark or most representative example of the stands used to develop the FBP System. The pole in the photos is 180 cm tall. Additional photographs showing typical variations in stand structure and composition (indicating the condition to which each fuel type is applicable) are found in the poster Groot (1993).

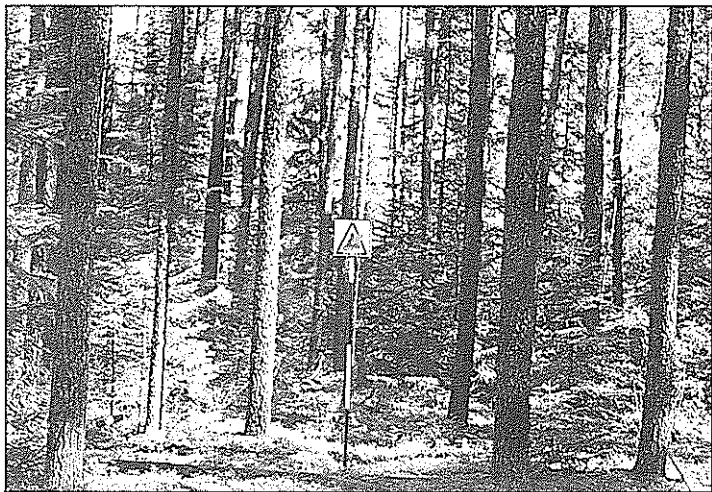
¹ from De Groot (1993).



C-1 Spruce-Lichen Woodland Open black spruce stands with lichen understory.



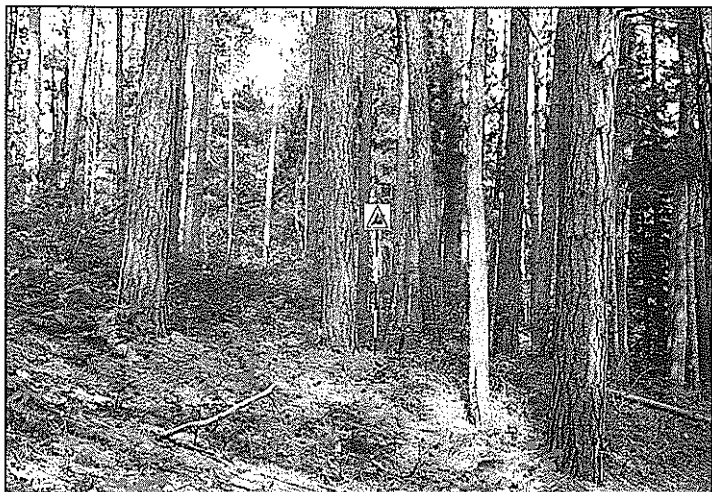
C-2 Boreal Spruce Upland and lowland black spruce, white and Engelmann spruce stands. Does not include spruce-sphagnum bogs. Photo shows upland black spruce.



C-3 Mature Jack or Lodgepole Pine Fully stocked mature jack and lodgepole pine stands. Photo is of a jackpine stand.



Immature Jack or Lodgepole Pine Stands Densely stocked immature jack and lodgepole pine stands. Photo is of a jackpine stand.



C-5 Red and White Pine Mature red and white pine.



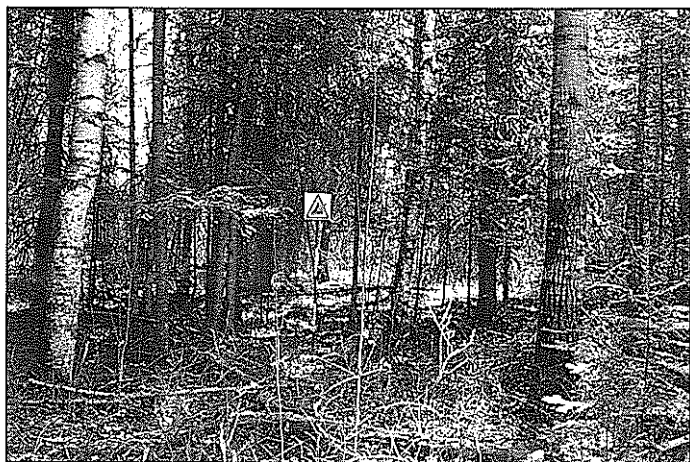
C-6 Conifer Plantation This fuel type is applicable to all conifer plantations with closed crown canopy and no understory shrub layer. Photo is of a red pine plantation.



C-7 Ponderosa Pine / Douglas-fir Mixed stands of uneven-aged ponderosa pine and Douglas-fir.



Leafless Aspen Pure semi-mature trembling aspen stands in the leafless stage.



M-1 Boreal Mixedwood - Leafless Mixed stands of boreal species in the leafless stage. Photo shows a stand with 75% coniferous and 25% deciduous component.



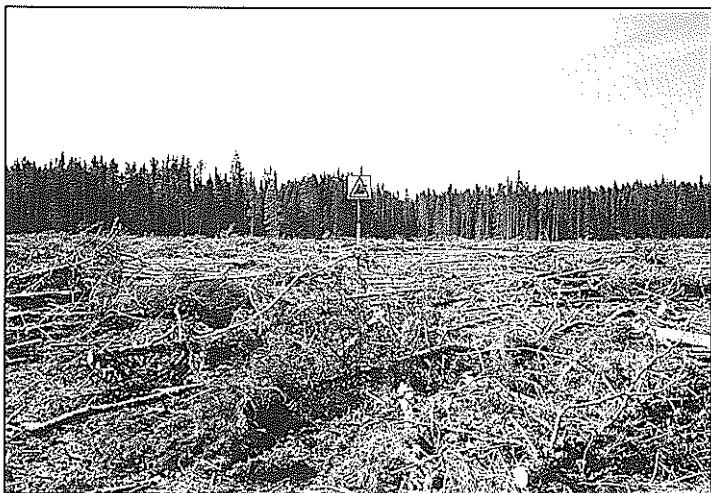
M-2 Boreal Mixedwood - Green Mixed stands of boreal species in the green stage. Photo shows a stand with 75% coniferous and 25% deciduous component.



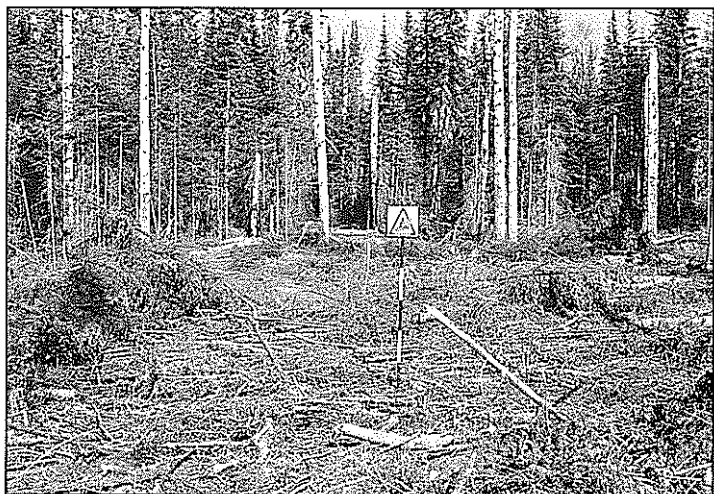
M-3 Dead Balsam Fir/Mixedwood - Leafless Mixed stands of dead balsam fir and boreal mixed wood species in the leafless stage. The stand in the above photo has 60% dead balsam fir and 40% live mixedwood species.



Dead Balsam Fir/Mixedwood - Green Mixed stands of dead balsam fir and boreal mixed wood species in the green stage. The stand in the above photo has 60% dead balsam fir and 40% live mixedwood species.



S-1 Jack or Lodgepole Pine Slash Jack or lodgepole pine slash, 1-2 seasons old.



S-2 White Spruce/Balsam Slash White or Engelmann spruce and balsam or subalpine fir slash, 1-2 seasons old.



S-3 Coastal Cedar/Hemlock/Douglas-fir Slash Western red cedar, western hemlock, and Douglas-fir slash, 1 season old. This fuel type is also applicable to interior wet belt cedar-hemlock slash.



grass Matted and standing grass. The above photo shows well-cured grass with a 3 t/ha fuel load.

Appendix 4 Beaufort scale of winds¹

Wind Speed (km/h)			
Range	Mean	Description	Observed Wind Effects
<2	0	Calm	Smoke rises vertically
2-5	3	Light Air	Direction of drift shown by smoke drift but not by wind vanes
6-11	9	Light Breeze	Wind felt on face, leaves rustle; vanes moved by wind
12-19	16	Gentle Breeze	Leaves and twigs in constant motion; wind extends light flag
20-29	24	Moderate Breeze	Raises dust and loose paper, small branches are moved
30-39	34	Fresh Breeze	Small trees in leaf begin to sway; crested wavelets on inland waters
40-50	44	Strong Breeze	Large branches in motion, whistling in telephone wires, umbrellas used with difficulty
51-61	55	Near Gale	Whole trees in motion; inconvenience felt when walking against wind
62-74	68	Moderate Gale	Breaks twigs off trees, generally impedes progress
75-87	82	Strong Gale	Slight structural damage occurs
88-101	96	Whole Gale	Seldom experienced inland, trees uprooted; considerable structural damage
102-116	110	Storm	Very rarely experienced; widespread damage
116+	125	Hurricane	

¹ Adapted from List (1951).

Appendix 5 FFMC adjustments¹

Tables A.1 and A.2 can be used to estimate the FFMC for each hour from noon of the current day to noon of the subsequent day for latitudes 48–60° N, in the absence of rain. Table A.3 can be used for topographic adjustments.

Table A.1 Diurnal FFMC Adjustments: 1200-0559 LST

Local Standard Time				Std. FFMC	Local Standard Time															
1200	1300	1400	1500	1600	1700	1800	1900	2000	2100	2200	2300	2400	0100	0200	0300	0400	0500	0559		
41	43	46	48	50	51	52	53	53	52	51	50	49	48	47	46	45	44	43		
48	52	55	57	60	61	62	62	62	61	59	58	56	55	54	52	51	50	49		
57	61	65	68	70	70	71	70	69	68	66	65	63	62	60	59	58	56	55		
59	63	67	70	72	72	72	72	71	69	68	66	65	63	62	61	59	58	57		
62	66	70	72	74	74	74	73	72	71	69	68	66	65	63	62	61	59	58		
63	67	71	73	75	75	75	74	73	72	70	69	67	66	64	63	61	60	59		
64	68	72	74	76	76	76	75	74	72	71	69	68	66	65	64	62	61	60		
66	69	73	75	77	77	77	76	75	73	72	70	69	67	66	64	63	62	60		
67	71	75	76	78	78	78	77	76	74	72	71	69	68	67	65	64	63	61		
69	72	76	78	79	79	78	77	76	75	73	72	70	69	67	66	65	63	62		
71	74	77	79	80	80	79	78	77	76	74	73	71	70	68	67	66	64	63		
74	76	79	80	81	81	80	79	78	77	75	73	72	71	69	68	66	65	64		
76	78	80	81	82	82	81	80	79	77	76	74	73	71	70	69	67	66	65		
78	80	81	82	83	83	82	81	80	78	77	75	74	72	71	70	68	67	66		
80	81	82	83	84	84	83	82	81	79	78	76	75	73	72	70	69	68	67		
82	82	83	84	85	85	84	83	82	80	79	77	76	74	73	71	70	69	68		
83	84	85	85	86	86	85	84	83	81	79	78	77	75	74	72	71	70	68		
84	85	86	86	87	87	86	85	83	82	80	79	78	76	75	73	72	71	70		
85	86	87	87	88	88	87	86	84	83	81	80	79	77	76	74	73	72	71		
86	87	88	89	89	89	88	87	85	84	82	81	80	78	77	75	74	73	72		
88	88	89	90	90	90	89	88	86	85	83	82	81	79	78	77	75	74	73		
89	89	90	91	91	91	90	89	87	86	84	83	82	80	79	78	76	75	74		
90	90	91	92	92	92	91	90	88	87	85	84	83	81	80	79	77	76	75		
91	91	92	93	93	93	92	91	89	88	86	85	84	82	81	80	79	77	76		
92	93	93	94	94	94	93	92	90	89	88	86	85	84	82	81	80	79	77		
93	94	94	95	95	95	94	93	91	90	89	87	86	85	83	82	81	80	79		
94	95	95	96	96	96	95	94	92	91	90	88	87	86	85	84	82	81	80		
95	96	96	97	97	97	96	95	93	92	91	90	88	87	86	85	84	83	81		
96	97	97	98	98	98	97	96	94	93	92	91	90	88	87	86	85	84	83		
97	98	98	99	99	99	98	97	95	94	93	92	91	90	88	87	86	85	84		
98	99	99	100	100	100	99	98	96	95	94	93	92	91	90	89	88	86	85		
1300	1400	1500	1600	1700	1800	1900	2000	2100	2200	2300	2400	0100	0200	0300	0400	0500	0600	0659		
Local Daylight Time					Local Daylight Time															

To estimate the FFMC during the afternoon or overnight, find the daily standard FFMC then move horizontally to the column that corresponds to the prediction time.

¹ Adapted from Lawson et al. (1996)

Table A.2 Diurnal FFMC Adjustments: 0600-1159 LST

Standard Daily FFMC

Time LST	RH	50	60	70	72	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	95	96	97	98	99	100
0600	<58	54	57	62	63	64	65	66	66	67	68	69	69	70	71	72	73	74	75	76	78	79	80	81	83	84	86	87	89	90	92	93
	68-87	48	53	58	60	61	62	63	63	64	65	66	66	67	68	69	70	71	72	73	74	75	76	77	78	80	81	82	84	85	87	88
	>87	43	49	55	57	58	59	60	60	61	62	63	64	65	66	67	67	68	69	71	72	73	74	75	76	77	79	80	81	83	84	85
0700	<58	56	60	64	65	67	67	68	69	69	70	71	72	72	73	74	75	76	77	78	79	80	81	83	84	85	87	88	90	91	93	94
	58-77	49	54	60	61	63	63	64	65	66	67	67	68	69	70	71	72	73	74	75	76	77	78	79	81	82	83	84	86	87	88	90
	>77	44	49	56	57	59	60	60	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	77	78	79	81	82	83	85	86	88
0800	<48	59	63	67	68	69	70	70	71	72	72	73	74	75	75	76	77	78	79	80	81	82	83	84	85	86	88	89	90	92	93	95
	48-67	50	55	61	63	64	65	66	67	68	68	69	70	71	72	73	74	75	76	77	78	79	81	82	83	84	85	87	88	89	90	91
	>67	45	50	57	58	60	61	61	62	63	64	65	66	67	68	69	70	71	72	73	74	76	77	78	79	81	82	84	85	87	88	90
0900	<43	64	67	71	72	73	74	74	75	75	76	77	77	78	79	80	80	81	82	83	84	85	86	87	88	89	90	91	93	94	95	96
	43-62	56	60	66	67	69	69	70	71	72	72	73	74	75	76	77	77	78	79	80	81	82	83	85	86	87	88	89	91	92	93	94
	>62	51	56	62	63	64	65	66	66	67	68	69	70	70	71	72	73	74	75	76	77	78	79	80	82	83	84	85	87	88	90	91
1000	<38	70	72	76	77	77	78	78	79	79	80	81	81	82	82	83	84	85	85	86	87	88	89	90	91	92	93	94	95	96	97	98
	38-57	62	66	71	72	73	74	75	75	76	77	77	78	79	79	80	81	82	83	84	85	86	87	88	89	90	91	92	93	94	96	97
	>57	57	61	67	68	69	70	70	71	72	72	73	74	74	75	76	77	77	78	79	80	81	82	83	84	85	86	87	88	89	91	92
1100	<35	76	78	80	81	82	82	83	84	84	85	85	86	86	87	87	88	89	89	90	90	91	91	92	92	93	93	94	95	96	97	98
	35-54	69	0	72	76	77	78	79	80	80	81	82	82	83	83	84	85	86	87	87	88	88	89	89	90	91	91	92	93	94	96	97
	>54.5	64	68	72	73	74	75	76	76	77	77	78	78	79	80	80	81	81	82	82	83	84	84	85	85	86	86	87	88	89	91	92
1159	<33	82	83	85	86	87	87	87	88	88	88	89	89	89	90	90	90	91	91	91	92	92	92	92	93	93	93	94	95	96	97	98
	33-52	77	79	82	83	84	84	84	85	85	86	86	86	87	87	88	88	88	89	89	89	90	90	90	91	91	91	92	93	94	96	97
	>52	72	75	78	79	80	80	81	81	81	82	82	82	83	83	83	83	84	84	84	85	85	85	86	86	86	86	87	88	89	90	92

To estimate the FFMC in the morning, find the previous day's standard FFMC, then move down to the time and relative humidity for the prediction time. If the RH is unknown use the highest value.

TABLE A.3 FFMC slope - aspect adjustments

Level	Ground Slope and Aspect															
	1 - 15 %				16 - 30 %				31 - 45 %				46 - 60 %			
	N	E	S	W	N	E	S	W	N	E	S	W	N	E	S	W
80	78	79	82	80	77	78	82	80	74	77	83	81	72	76	84	81
82	80	81	84	82	79	80	84	82	76	79	85	83	74	78	85	83
84	83	83	85	84	81	82	86	84	79	81	87	84	76	80	88	84
86	85	85	87	86	83	84	88	86	81	83	89	86	78	82	90	86
87	86	86	88	87	84	85	89	87	82	84	90	87	80	83	90	87
88	87	87	89	88	85	87	90	88	83	86	91	88	82	85	91	88
89	88	88	90	89	87	88	91	89	85	87	91	89	83	86	92	89
90	89	89	91	90	88	89	92	90	86	88	92	90	84	87	93	90
91	90	90	92	91	89	90	92	91	87	89	93	91	86	88	93	91
92	91	91	93	92	90	91	93	92	88	90	94	92	87	89	94	92
93	92	92	94	93	91	92	94	93	89	91	95	93	88	90	95	93
94	93	93	95	94	92	93	95	94	91	92	96	94	90	92	96	94

Determine the slope and aspect of the weather observation point, find FFMC, then move horizontally to the column that best describes the prediction point and read **adjusted FFMC** (from Alexander et al. 1984). These adjustments should be used with caution as they have not been rigorously tested. They should only be applied in slash and open fuel types on clear days in March, April, August, September or October between 1200 -2000 LST.

Appendix 6 Fire Behavior Prediction Worksheet Example

Fire No./Name	SAMPLE	Date	29.04.95	Time	1600 LST
Prediction Date & Time	30.04.95	From	1200 LST	To	1300 LST
Prediction Point	1 PI	2 PI	3 APS	4 APS	5 PI
Fuel Type					
FBP System fuel type	C-4	M-1	D-1	S-2	O-1a
Modifiers (CBH,C:H,PDF,%Cured)		75%			95%
Fine Fuel Moisture Code					
Standard daily FFMC	93	94	88	94	94
Diurnal or Hourly FFMC	91	92	85	92	92
Ground slope (%)	0	15	10	50(15)	20(L)
Aspect	-	N	E	S(E)	W(L)
Adjusted FFMC	91	92	85	95	92
Initial Spread Index					
10 m wind speed (km/h)	23	15	-3	5	-15
Slope equiv wind speed (km/h)	-	5	3	26	8
Effective wind speed (km/h)	23	20	0	31	-7
ISI	16	16	2	41	8
BISI	2	2	2	-	4
Rate of Spread and Intensity					
BUI	100	25	60	100	NA
Equil. head fire ROS (m/min)	27	16	0.2	-	9
Back fire ROS (m/min)	2	2	0.2	-	9
Fire intensity class	6	5	2	6	3
Type of fire	C	I	S	S	S
Fire Size					
Elapsed time (min)	60	60	60	60	60
Head fire spread dist. (m)	1362	757	12	1980	1077
Back fire spread dist. (m)	101	50	-	-	462
Total spread dist. (m)	1462	807	12	1980	1539
Fire area (ha)	62	20			65
Perimeter length (m)	3350	1830			3400
Perimeter growth rate (m/min)	61	41			44
Remarks					

Appendix 6 Fire Behavior Prediction Worksheet

The attached worksheet may be used to record the input data required for the procedures in the section Use of the Guide and the resulting fire characteristics (see the sample worksheet).

1. Heading

Enter Fire Number and or Name; Date and Time; Prediction Date and Time Interval; Prediction Point. All calculations made on one sheet should be for the same time interval. Use a new sheet for each successive or new time interval. Use a separate prediction point for each different fuel type or slope/aspect class that is likely to be encountered during the prediction time interval. Specify the type of ignition at each prediction point: point ignition (P.I.) or active perimeter source (A.P.S.).

2. Fuel Type

Enter the Fuel Type Identifier which best represents the fuels at the prediction point. Also record any relevant modifiers (C-6 enter Crown Base Height; M-1 and M-2 enter Percent Softwood/Hardwood; M-3 and M-4 enter Percent Dead Fir; O-1 enter Percent Cured).

3. Fine Fuel Moisture Code

Enter the hourly FFMC for the Prediction Date/Time if known, or enter the Standard Daily FFMC for the Prediction Date and use it to determine the diurnal FFMC for the Prediction Time. Enter the Ground Slope (%) and Aspect (cardinal direction) for the area ahead of the Prediction Point, disregarding minor variations. If topographic adjustment is required, also record the slope and aspect of the weather observation point in brackets (use L for level): e.g. Ground slope 50 [15]; Aspect S [E]. Use these values to determine the Adjusted FFMC.

4. Initial Spread Index

Enter the 10 m open Wind Speed for the Prediction Time as a positive value if the wind is blowing upslope, or a negative value if it is blowing downslope. Enter the estimated values of Slope Equivalent Wind Speed, Effective Wind Speed (Open WS + Slope Equivalent WS), ISI, and BISI. ISI and BISI values are wind and slope adjusted.

5. Rate of Spread and Intensity

Enter BUI for the Prediction Date and record the Head Fire ROS, Back Fire ROS, Fire Intensity Class and Type of Fire from the tables.

Record Type of Fire as S (surface fire), I (intermittent crown fire), or C (continuous crown fire).

6. Fire Size

Enter the Elapsed Time corresponding to the Prediction Time interval.

Record the predicted Head fire Spread Dist., Back fire Spread Dist., Total Spread Dist. (Head fire + Back fire Spread Dists.), Fire Area, Perimeter Length, and Perimeter Growth Rate.

7. Remarks

Record any fire characteristics observed at the prediction time (Head fire ROS, Spread Dist., Type of Fire).

A blank form that may be reproduced is given on the next page.

FIRE BEHAVIOR PREDICTION

Fire No./Name		Date		Time	
Prediction Date & Time		From		To	
Prediction Point	1	2	3	4	5
Fuel Type					
FBP System fuel type					
Modifiers (CBH,C:H,PDF,%Cured)					
Fine Fuel Moisture Code					
Standard daily FFMC					
Diurnal or Hourly FFMC					
Ground slope (%)					
Aspect					
Adjusted FFMC					
Initial Spread Index					
10 m wind speed (km/h)					
Slope equiv wind speed (km/h)					
Effective wind speed (km/h)					
ISI					
BISI					
Rate of Spread and Intensity					
BUI					
Equil. head fire ROS (m/min)					
Back fire ROS (m/min)					
Fire intensity class					
Type of fire					
Fire Size					
Elapsed time (min)					
Head fire spread dist. (m)					
Back fire spread dist. (m)					
Total spread dist. (m)					
Fire area (ha)					
Perimeter length (m)					
Perimeter growth rate (m/min)					
Remarks					

Probabilities of Sustained Ignition in Lodgepole Pine, Interior Douglas-fir, and White Spruce- Subalpine Fir Forest Types

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Canadian Forest Service
Natural Resources Canada
Victoria, B.C.

October 1996

Supplement 1 to:
Field guide to the Canadian
Forest Fire Behavior Prediction (FBP) System.
FRDA Handbook
012

Partnership Agreement on Forest Resource Development: FRDA II

Canada



Introduction

Tables of sustained ignition probabilities are presented for four benchmark forest fuel types: dry- and moist-site lodgepole pine, white spruce-subalpine fir, and interior Douglas-fir. The look-up tables were derived from logistic regression equations that were fit to hundreds of small-scale match-fire and campfire observations made near Prince George and 100 Mile House, B.C. by Canadian Forest Service fire researchers. The field research studies were carried out in the 1950s and 1960s as part of the early development of the Canadian Forest Fire Danger Rating System, but the original data have been re-analyzed using current versions of components of the Canadian Forest Fire Weather Index System as predictors of ignition probability.

A complete scientific and technical description of the development of the lodgepole pine and spruce-subalpine fir equations is given in Lawson *et al.* (1994a), and is illustrated in a wall poster (Lawson *et al.* 1994b) as graphs with colour photos of the forest types. A computer application (Lawson *et al.* 1996) links these same equations with a new diurnal Fine Fuel Moisture Code program (Lawson, Armitage and Hoskins 1996). Derivation of the interior Douglas-fir equation is presented in the Appendix.

The following four equations are used to calculate the ignition probabilities presented in tables S1.1-S1.4:

Dry lodgepole pine

$$P = 1 / (1 + \exp(2.107 - 0.727 * ISI)) \dots\dots\dots S1.1$$

Moist lodgepole pine

$$P = 1 / (1 + \exp(2.146 - 0.009 * BUI - 0.349 * ISI)) \dots S1.2$$

White Spruce - subalpine fir

$$P = 1 / (1 + \exp(2.766 - 0.005 * DC - 0.396 * ISI)) \dots S1.3$$

Interior Douglas-fir

$$P = 1 / (1 + \exp(1.563 - 0.005 * BUI - 0.478 * ISI)) \dots S1.4$$

Probability of sustained ignition is not a standard output from the Canadian Forest Fire Behavior Prediction (FBP) System (Forestry Canada Fire Danger

Group 1992), but is included here as a supplement for the convenience of users. Potential uses include gauging the likelihood of fire starts or fire arrivals for preparedness planning of initial attack crew positioning and readiness, and judging spot-fire ignition potential for wildfires or prescribed fires. The ignition probabilities described here relate to the probability of an ignition from small fire brands, such as matches or camp fires persisting with flaming combustion (for at least 2 minutes for match fires and 15 minutes for campfires) and reaching the threshold condition for fire spread and growth. Camp fires were necessary for sustained ignition in the white spruce-subalpine fir forest type. Matches and camp fires were both potentially successful fire brands in the other forest types, depending on burning conditions.

Use of the Tables

The following steps form the procedure for using the Ignition Probability Tables (Table and Appendix numbers refer to Taylor *et al.* 1996).

1. Select the most appropriate forest type and/or variant from the four presented.

Detailed forest type descriptions and a photograph accompany each ignition probability table. Ignition probability differences among variants within a single FBP System fuel type should be expected. The “dry” and “moist” lodgepole pine forest types can be considered variants of C-3 (Mature Jack or Lodgepole Pine). The interior Douglas-fir forest type should be considered a variant of C-7 (Ponderosa pine - Douglas-fir). The white spruce-subalpine fir forest type can be considered a variant of C-2 (Boreal spruce), as shown by DeGroot (1993).

2. Determine the most representative Fine Fuel Moisture Code (FFMC), 10-m open wind speed, and Initial Spread Index (ISI) for the time of interest.
 - a) The hourly FFMC (Van Wagner 1977) should be used, if available. Alternatively, the diurnal FFMC (Tables A.1 - A.2, Appendix 5) may be used for any desired time of the day. The daily standard FFMC may be used to represent mid-afternoon (16:00 hr LST) conditions.

- b) Wind speed at 10 m in the open should be measured, forecasted, or estimated from the Beaufort Scale (Appendix 4) for the time of interest.
- c) For the "Dry" Lodgepole Pine forest type, enter Table S1.1 with FPMC and 10-m wind speed from 2a and 2b, above.

For other forest types and/or variants, determine the ISI using Table 3, with FPMC and 10-m open wind speed from 2a and 2b, above.

- 3. Determine the Buildup Index (BUI) or Drought Code (DC) from the nearest representative weather station, as required for a given forest type.
- 4. From the ignition probability table for the forest type of interest, find the probability-of-ignition value (%) at the intersection of the two required input values. Probability-of-ignition classes (low, medium, high) are differentially shaded in the body of each table.

References

- De Groot, W.J. 1993. Examples of fuel types in the Canadian Forest Fire Behavior Prediction System. For. Can., Nor. For. Cent., Edmonton, Alta. Poster (with text).
- Forestry Canada Fire Danger Group. 1992. Development and structure of the Canadian Forest Fire Behavior Prediction System. For. Can., Inf. Rep. ST-X-3, Ottawa, Ont. 64 p.
- Lawson, B.D., O.B. Armitage, and G.N. Dalrymple. 1994a. Ignition probabilities for simulated people-caused fires in B.C.'s lodgepole pine and white spruce-subalpine fir forests. *In* Proc. 12th Conf. on Fire and Forest Meteorology; Oct. 26-29, 1993, Jekyll Is., Ga., Soc. Am. Foresters, Bethesda, Md. SAF Publ. 94-02. pp. 493-505.
- _____. 1994b. Ignition probabilities for lodgepole pine and spruce-subalpine fir forests. Can. For. Serv. and B.C. Min. of For., Victoria, B.C. FRDA Poster (with text).
- _____. 1996. Wildfire Ignition Probability Predictor (WIPP). R&D Update. Can. For. Serv., Pac. For. Cent., Victoria, B.C., 2 p.
- Lawson, B.D., O.B. Armitage, and W.D. Hoskins. 1996. Diurnal variation in the Fine Fuel Moisture Code: tables and computer source code. Can. For. Serv. and B.C. Min. For., Victoria, B.C., FRDA Rep. No. 245, 20 p.
- Taylor, S.W., R.G. Pike, and M.E. Alexander. 1996. Field guide to the Canadian Forest Fire Behavior Prediction (FBP) System. Can. For. Serv. and B.C. Min. For., Victoria, B.C. FRDA Handb. No. 012.
- Van Wagner, C.E. 1977. A method of computing fine fuel moisture content throughout the diurnal cycle. Can. For. Serv., Petawawa For. Exp. Sta., Chalk R., Ont. Inf. Rep. PS-X-69.

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“Dry” lodgepole pine forest-type characteristics

- * Pure, moderately dense stands of mature lodgepole pine;
- * scattered pine understory;
- * sandy well-drained soils, low-productivity sites dominated by feather-moss and *Cladina* (lichen); B.C. biogeoclimatic classification SBSmk1/03 is typical;
- * forest floor shallow (3-5 cm), low bulk density; and
- * fire-carrying fuels - lichen, bearberry (*Arctostaphylos uva-ursi*), feathermoss (*Pleurozium schreberi*; *Dicranum* spp.), needle litter.



Table S1.1

“Dry” lodgepole pine forest type - probability of sustained ignition (%) and ignition class

FFMC	10-m Open Wind Speed (km/h)										
	0	5	10	15	20	25	30	35	40	45	≥50
70	16	18	21	24	30	38	49	63	79	91	97
72	17	19	21	26	32	40	53	68	82	93	98
74	17	19	23	27	34	44	57	73	87	95	99
76	18	21	24	30	38	49	64	79	91	97	99
78	19	23	27	34	44	58	73	87	95	99	100
80	22	26	32	41	54	69	84	94	98	100	100
82	26	32	40	52	68	82	93	98	100	100	100
84	32	40	53	68	83	93	98	100	100	100	100
86	41	54	69	84	94	98	100	100	100	100	100
88	56	71	85	95	99	100	100	100	100	100	100
90	73	87	95	99	100	100	100	100	100	100	100
92	88	96	99	100	100	100	100	100	100	100	100
94	97	99	100	100	100	100	100	100	100	100	100
96	99	100	100	100	100	100	100	100	100	100	100
≥98	100	100	100	100	100	100	100	100	100	100	100

Ignition Class		Probability %
	LOW	0-49
	MEDIUM	50-75
	HIGH	76-100

75

“Moist” lodgepole pine forest-type characteristics

- * Mixed, well-stocked stands of mature lodgepole pine-Douglas-fir;
- * significant understory of lodgepole pine, trembling aspen, white spruce, and subalpine fir;
- * moderately well-drained soils typically derived from unstratified glacial deposits, medium site productivity, *Cornus canadensis* (bunchberry), scattered *Cladonia* (lichen), feather mosses, and *Vaccinium membranaceum* (black huckleberry); B.C. biogeoclimatic classification SBSmk1/05 is typical;
- * forest floor shallow (3-5 cm), low bulk density; and
- * fire-carrying fuels - feather moss, needle litter.



Table S1.2

“Moist” lodgepole pine forest type - probability of sustained igniton (%) and ignition class.

	Buildup Index (BUI)							
	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
ISI								
0.5	13	15	16	18	21	26	33	41
1	15	17	19	21	24	29	37	46
1.5	18	20	21	24	27	33	41	50
2	20	23	24	27	31	37	45	54
2.5	23	26	28	31	34	41	50	59
3	27	29	31	34	38	45	54	63
4	34	37	39	43	47	54	62	70
5	42	46	48	51	56	62	70	77
6	51	54	57	60	64	70	77	83
7	60	63	65	68	72	77	83	87
8	68	70	72	75	78	82	87	91
9	75	77	79	81	84	87	91	93
10	81	83	84	86	88	90	93	95
11	86	87	88	90	91	93	95	96
12	89	91	91	92	94	95	96	97
13	92	93	94	94	95	96	97	98
14	94	95	95	96	97	97	98	99
15	96	96	97	97	98	98	99	99
18	99	99	99	99	99	99	100	100

	Ignition Class	Probability %
	LOW	0-49
	MEDIUM	50-75
	HIGH	76-100

White spruce-subalpine fir forest-type characteristics

- * Mixed, moderately dense stands of mature white (and/or Engelmann) spruce-subalpine fir, with scattered mature birch and Douglas-fir;
- * significant understory of subalpine fir;
- * soils range from loamy sands to sticky, poorly drained clays on moderately well-drained to imperfectly drained sites; herb and shrub layers dominated by *Aralia nudicaulis* (sarsaparilla), *Gymnocarpium dryopteris* (oak fern), *Vaccinium* (huckleberry) and *Cornus* (bunchberry) spp., *Oplopanax horridus* (devil's club) on wettest sites; mosses patchy; B.C. biogeoclimatic classification SBSmk1/07 (drier) and SBSmk1/08 (wetter) are typical;
- * forest floors moderate to deep (6-12 cm), moderate to high bulk density; and
- * fire-carrying fuels - litter, feather moss.



Table S1.3

White spruce - subalpine fir forest type - probability of sustained ignition (%) and ignition class

ISI	Drought Code (DC)							
	0 100	101 200	201 300	301 400	401 500	501 600	601 700	701 800
0.5	9	14	21	31	42	55	66	77
1	11	17	25	35	47	59	71	80
1.5	13	19	28	40	52	64	75	83
2	15	23	33	44	57	68	78	86
2.5	18	26	37	49	62	73	81	88
3	21	30	42	54	66	76	84	90
4	28	39	52	64	74	83	89	93
5	37	49	61	72	81	88	92	95
6	47	59	70	80	87	91	95	97
7	56	68	78	85	91	94	96	98
8	66	76	84	90	93	96	97	98
9	74	82	89	93	95	97	98	99
10	81	87	92	95	97	98	99	99
11	86	91	94	97	98	99	99	100
12	90	94	96	98	99	99	99	100
13	93	96	97	98	99	99	100	100
14	95	97	98	99	99	100	100	100
15	97	98	99	99	100	100	100	100
18	99	99	100	100	100	100	100	100

Ignition Class		Probability %
	LOW	0-49
	MEDIUM	50-75
	HIGH	76-100

Interior Douglas-fir forest-type characteristics

- * Well-stocked stands of interior Douglas-fir of varying age classes, including large, mature overstory trees with a significant understory;
- * forest floor shallow (less than 3 cm); and
- * fire-carrying fuels - grass covers the ground except under fir thickets, where needle litter and patchy moss predominates. Grass is assumed to be greater than 50% cured (dead), although the degree of curing is not a variable used in the ignition probability model.

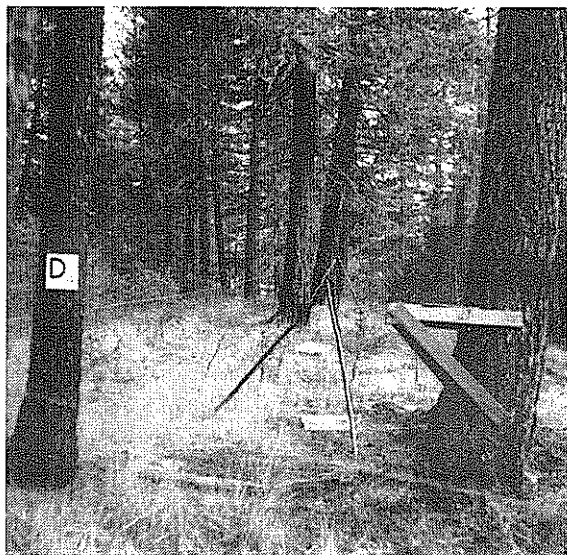


Table S1.4

Interior Douglas-fir - probability of sustained ignition (%) and ignition class

ISI	Buildup Index (BUI)							
	0 20	21 30	31 40	41 60	61 80	81 120	121 160	161 200
0.5	22	23	24	25	27	30	35	40
1	26	28	29	30	32	36	40	45
1.5	31	33	34	36	38	41	46	51
2	36	38	39	41	44	47	52	57
2.5	42	44	45	47	50	53	58	63
3	48	50	51	53	56	59	64	68
4	60	62	63	65	67	70	74	78
5	71	72	73	75	76	79	82	85
6	79	81	81	83	84	86	88	90
7	86	87	88	88	89	91	92	94
8	91	92	92	92	93	94	95	96
9	94	95	95	95	96	96	97	97
10	96	97	97	97	97	98	98	98
11	98	98	98	98	98	99	99	99
12	99	99	99	99	99	99	99	99
13	99	99	99	99	99	99	100	100
14	99	99	100	100	100	100	100	100
15	100	100	100	100	100	100	100	100

Ignition Class		Probability %
	LOW	0-49
	MEDIUM	50-75
	HIGH	76-100

APPENDIX

Derivation of the Interior Douglas-fir Ignition Probability Table S1.4

Background

The equation for probability of sustained ignition used to produce Table S1.4 was derived from analysis of a data set collected during the years 1957-1959 near 100 Mile House, B.C. by Dr. P.M. Paul, now retired research scientist, Canadian Forest Service. The test-fire program there resulted in publication of the Cariboo, British Columbia, Forest Fire Danger Tables in 1965, while the study sites themselves were described in detail in annual progress reports,¹ which were used for decisions about data selection for the present analysis and for the site photograph.

The data for site BC-801 were obtained from a national data base on test fires conducted in Canada between 1931-1961 as part of a Canadian forest Service program of regional forest fire danger table development, which preceded the present national fire danger rating system. The test fire data base, consisting of some 20 000 fires, has been edited, merged with current version FWI System component values, and assembled in modern computer formats, as described by T. Lynham.²

¹ Paul, P.M. 1957, 1958, 1959. Progress report on forest fire research, 100 Mile House, B.C., Can. For. Serv., For. Fire Res. Inst., File No. 668. Ottawa, Ont.

² Lynham, T.J. and D.L. Martell. 1989. Preliminary report on a national database of experimental fires in Canada. P41-44 in: Proc. National Workshop on forest fire occurrence prediction, May 3-4, 1989, Winnipeg, Man.

Lynham, T.J. 1992. Summary of data manipulations to Forestry Canada's 20 000 test-fire data base for addition of Can. FWI values. Rep. On file, GLFC, Sault Ste. Marie, Ont.

Analysis of 100 Mile House data

Data analysis followed the same steps described by Lawson *et al.* (1994a). Two suitable study sites were chosen for characterization as “interior Douglas-fir” from the nine available; specifically, forest type “D-1, Fir-grass/ moss” (site code 80104) and “D-4, Fir- grass/ needle” (site 80107) (photo of D-4 included with Table S1.4). Site descriptions are included here as they were presented in Paul (1957):

Site D1 – Douglas-fir in varying age classes. Large mature fir predominates, but there is an abundance of older immature (81-120 years) with well-established young growth in understory... Intermingled with the grass, which is ubiquitous except under the trees, are patches of moss...

Site D4 – is located on the southwest slope of the ridge at a slightly lower elevation than D1, which occurs on a bench near the summit of the hill. It is well stocked, with the fir trees attaining a very good size, although smaller than the mature and overmature trees of D1. Young reproduction, up to 4 feet, is extensive, and there is a fairly complete ground cover of grass, although it is continuously broken by patches of fir needles.

Paul (1957, 1958, and 1959) reported progression of grass curing through the test-fire seasons. Curing started in late May 1957 as 75-80%, decreasing to 35- 45 % by mid- July, and increasing again during September to about 70 % by late September. However, in 1958 (a drier summer) grass was 75-85% cured by late August. Grass curing could not be analyzed as an independent variable in the model, but an assumption could be made that test fires were usually conducted when grass was 50% or more cured.

Before the test-fire data were analyzed for the two sites, diurnally adjusted FFMC and ISI were calculated for each test fire using the diurnal FFMC program from Lawson, Armitage and Hoskins (1996) and the noon 10-m open wind speed recorded at the headquarters weather station. A dependent binary variable, “ignition”, was defined as “yes” if the burned area for match fires was more than 0.30 ft², and “no” if less than or equal to 0.30 ft². This area criterion was chosen arbitrarily after inspecting the data and comparing them with results from a separate analysis on “vigor” class (on file, Pacific Forestry Centre). While the burned-area threshold for successful ignition

was larger than the 0.05 ft² used by Lawson *et al.* (1994a) for lodgepole pine match fires, the interior Douglas-fir forest types were faster spreading, being grass dominated.

Logistic regression was used to develop a preliminary model of "ignition" as predicted from diurnally adjusted FFMC, DMC, DC, ISI, and BUI. A total of 391 match fires (no camp fires) were available from sites D-1 and D-4 combined, with all D-1 fires recording an area greater than 0, while 80% of D-4 fires burned an area greater than 0.

A logistic regression equation that included ISI and BUI produced the statistically best results from the two independent variables (adding DMC increased the r^2 and total correct statistics slightly). The equation for probability of sustained ignition in interior Douglas-fir is

$$P = 1 / (1 + \exp (1.563 - 0.005 * \text{BUI} - 0.478 \text{ ISI}))$$

The equation is plotted in three dimensions in Figure 1, and Table 1 summarizes the observed and expected outcomes by decile and overall goodness-of-fit.

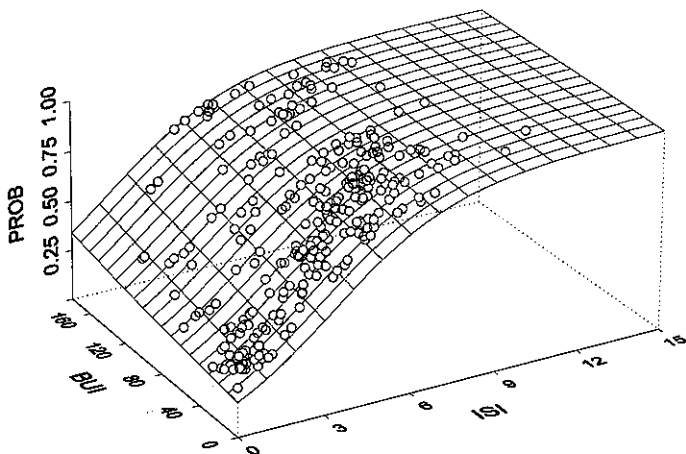


Figure 1. Probability of sustained ignition vs. Initial Spread Index (ISI) and Buildup Index (BUI) for interior Douglas-fir forest type. Plotted points represent test fires from sites D1 and D4, 100 Mile House, B.C.

Table 1. Logistic regression statistics: observed and expected outcomes by decile, overall goodness-of-fit, and success of prediction by the model, interior Douglas-fir forest type.

P	P	Total	<u>Ignition</u>		<u>No Ignition</u>		χ^2	P-value	R ²
Upper Limit	Avg.	Fires	Obs.	Exp	Obs.	Exp	(8 df)		
0.307	0.262	38	13	10.0	25	28.0			
0.414	0.350	39	17	13.7	22	25.3			
0.589	0.509	40	16	20.4	24	19.6			
0.657	0.616	39	18	24.0	21	15.0			
0.743	0.704	39	27	27.5	12	11.5			
0.790	0.766	38	29	29.1	9	8.9			
0.843	0.820	40	36	32.8	4	7.2			
0.883	0.860	39	36	33.5	3	5.5			
0.923	0.902	38	35	34.3	3	3.7			
1.000	0.947	41	37	38.8	4	2.2			
							13.1	0.106	0.189
391									

Success of prediction by the model

	Ignition Predicted	No Ignition Predicted	Total
Ignition	197.6	66.4	264
No Ignition	66.4	60.6	127
Predicted Total	264.0	127.0	391
Proportion Correct	0.749	0.477	
Total Correct	0.661		
Success Index	0.073	0.153	